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RESEARCH ARTICLE

Plasma extracellular vesicles from *APOE3* Christchurch carriers display a protective phenotype in early stages of autosomal dominant Alzheimer's disease

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Abstract

INTRODUCTION: The *PSEN1*^{E280A} mutation causes autosomal dominant Alzheimer's disease (ADAD) with predictable onset, enabling presymptomatic studies. Extracellular vesicles (EVs) are emerging biomarkers of cognitive decline, but their role in early ADAD is unclear. The rare apolipoprotein E (*APOE3*) Christchurch (*APOE3*^{Ch}) variant delays disease onset, yet its effect on EVs is unknown.

METHODS: We analyzed plasma EVs from mild cognitive impairment (MCI) and non-MCI *PSEN1*^{E280A}-*APOE3* carriers and non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch} carriers using flow cytometry, proteomics, and co-culture assays.

RESULTS: *APOE3*^{Ch}-EVs showed reduced vascular activation and inflammatory cargo linked to β-catenin signaling, higher apoE levels, and enrichment in lipid-loaded EVs. They mimicked the protective effect of recombinant ApoE3Ch on endothelial integrity by restoring β-catenin nuclear localization. In contrast, EVs from non-MCI *PSEN1*^{E280A}-*APOE3* carriers displayed vascular and inflammatory signatures associated with poorer cognition and detrimental astrocyte–endothelium effects. These findings highlight *APOE3*^{Ch}-EVs as modulators of vascular and inflammatory pathways with biomarker and therapeutic potential in ADAD.

Lina Pineda-Lopez and David Aguillon contributed equally to this study.

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KEYWORDS
apolipoprotein E ε3 Christchurch, autosomal dominant Alzheimer's disease, cognitive decline, extracellular vesicles, inflammation, presenilin1-E280A

Highlights

- Plasma extracellular vesicles (EVs) from non-mild cognitive impairment (MCI) *PSEN1*^{E280A}-apolipoprotein E (*APOE3*) carriers showed vascular and inflammatory signatures correlating with poorer cognition and detrimental effects on gliovascular co-cultures.
- *PSEN1*^{E280A}-*APOE3*^{Ch} EVs displayed reduced vascular and leukocyte markers, enrichment in apoE and lipid-loaded EVs, and proteomic signatures opposite to those of *PSEN1*^{E280A}-*APOE3* EVs.
- Functional assays revealed that *PSEN1*^{E280A}-*APOE3*^{Ch} EVs preserved endothelial barrier integrity and reduced astrocytic reactivity, partially linked to modulation of β-catenin signaling.
- Pathway analysis indicated that *APOE3*^{Ch}-EVs suppressed inflammatory, vascular, and coagulation pathways.
- These findings identify EVs as early biomarkers of disease progression and resilience in autosomal dominant Alzheimer's disease.

1 | BACKGROUND

Autosomal dominant Alzheimer's disease (ADAD) is an inherited form of Alzheimer's disease (AD), the leading cause of dementia worldwide.[1] ADAD results from mutations in the amyloid precursor protein (*APP*) gene or in the presenilin genes *PSEN1* and *PSEN2*, which are critical for APP processing.[2] Among these, the *PSEN1*^{E280A} variant causes an aggressive form of ADAD, with carriers typically developing cognitive decline before age 47.[3] The Paisa kindred in Colombia, which harbors this mutation, is the largest known family affected by ADAD due to *PSEN1*^{E280A}.[4] Within this cohort, a remarkable case of resistance to ADAD was identified: a woman homozygous for the rare apolipoprotein E (*APOE3*) Christchurch (*APOE3*^{Ch}) variant (R136S) who remained cognitively unimpaired until her seventies, despite carrying the *PSEN1*^{E280A} mutation.[5] Beyond this homozygous case, recent evidence indicates that heterozygous *APOE3*^{Ch} carriers exhibited a median 5-year delay in the onset of mild cognitive impairment (MCI) compared to non-carriers. Neuroimaging data further revealed relatively preserved cortical structure with low tau accumulation in regions typically affected by early neurodegeneration.[3]

Mechanistic studies demonstrate that the *APOE3*^{Ch} variant reduces binding to heparan sulfate proteoglycans (HSPGs), promoting cellular uptake and tau degradation and reducing tauopathy propagation.[5,6] It also displays reduced affinity for the low-density lipoprotein receptor (LDLR), decreasing lipid uptake and intracellular accumulation of toxic lipids,[7] and enhances Wnt/β-catenin signaling, a pathway critical for neuronal homeostasis.[8] These mechanisms are closely linked to neurovascular unit (NVU) integrity, which depends on endothelial cells, astrocytes, microglia, and mural cells, and whose dysfunction contributes to AD pathogenesis.[9,10] *Post mortem* studies revealed that the brain of a homozygous *APOE3*^{Ch} carrier exhibited minimal astrocytic and microglial activation and clear vascular protection, while heterozygous carriers showed reduced cerebral amyloid angiopathy, attenuated parenchymal vascular abnormalities, and less microvessel fragmentation compared to typical *PSEN1*^{E280A} cases.[3,11,12]

Peripheral biomarkers are increasingly recognized as valuable for detecting early brain changes in MCI and AD, supporting diagnostic and therapeutic strategies at preclinical stages.[13] In non-demented individuals, lipid, inflammatory, and metabolic profiles predict cognitive performance and its trajectory.[14,15] In the Paisa kindred, plasma alterations such as elevated amyloid beta $(A\beta)_{1\text{-}42}$, altered $A\beta_{1\text{-}42}/A\beta_{1\text{-}40}$ ratios, and increased tau phosphorylated at threonine 217 (p-tau217) appear before clinical symptom onset.[16,17] However, these biomarkers mainly reflect downstream pathology and provide limited insight into upstream mechanisms driving disease onset and progression. As NVU dysfunction, a key regulator of brain–periphery communication, is implicated in AD pathogenesis, biomarkers capturing early NVU alterations, including those from systemic or peripheral factors, are needed.

In the peripheral circulation, extracellular vesicles (EVs) and lipoproteins are the predominant nanoparticle populations.[18] *APOE3*^{Ch} carriers are known to have hyperlipoproteinemia[5] though it is unknown whether they have differences in the EVs properties and functional roles. EVs are nanoscale lipid bilayer-enclosed structures

RESEARCH IN CONTEXT

1. **Systematic review**: Previous studies have shown that extracellular vesicles (EVs) can propagate tau pathology and reflect neuroinflammatory and vascular processes in Alzheimer's disease (AD). However, their role in capturing early changes in autosomal dominant AD (ADAD) and in mediating protection conferred by apolipoprotein E (*APOE3*) Christchurch ($APOE3^{Ch}$) remains unclear.
2. **Interpretation**: We provide the first characterization of plasma EVs in cognitively unimpaired $APOE3^{Ch}$ carriers, with or without the $PSEN1^{E280A}$ mutation, compared to $PSEN1^{E280A}$-*APOE3* carriers stratified by cognitive status. Our results show that $APOE3^{Ch}$ reshapes EV composition toward a phenotype enriched in apoE and neutral lipids, while dampening vascular and inflammatory signals and preserving gliovascular integrity. These findings offer mechanistic insight into $APOE3^{Ch}$-associated resilience and highlight EVs as sensitive indicators of early disease processes.
3. **Future directions**: Further longitudinal and lipidomic studies are needed to determine whether $APOE3^{Ch}$-EV profiles predict delayed clinical onset, and to test whether modulating EV content can be leveraged for biomarker discovery or therapeutic interventions aimed at preserving neurovascular function in AD.

secreted by cells that carry proteins, lipids, and nucleic acids.[19] They mediate intercellular communication and reflect the physiological or pathological state of their origin.[20] In AD, EVs contribute to tau propagation, neuroinflammatory signaling, astrocyte activation, and vascular dysfunction.[21–25] Altered EV profiles are detectable at prodromal stages such as MCI, supporting their value as early biomarkers.[26,27] To investigate their role in early ADAD, we analyzed EV origin and protein composition in $PSEN1^{E280A}$-*APOE3* carriers, comparing non-MCI and MCI individuals, and explored potential protective mechanisms in non-MCI $PSEN1^{E280A}$-$APOE3^{Ch}$ carriers, a group informative for clinical resistance.

APOE isoforms influence EV biogenesis and cargo through effects on lipid membrane dynamics and endosomal trafficking.[28,29] Because the Christchurch variant alters lipid binding and receptor interactions, key processes in lipid metabolism[7] and EV biogenesis, it may shape EV content in ways that confer protection. Understanding how EVs reflect or mediate $APOE3^{Ch}$-associated protection could guide biomarker discovery and novel therapies targeting early neurovascular dysfunction in AD.

2 | METHODS

2.1 | Human samples

The study was approved by the University of Antioquia's review boards (Record No. 119 dated August 14, 2018- Sede de Investigación Universitaria, Universidad de Antioquia), and all participants gave written informed consent. The study included five groups of participants: cognitively unimpaired controls (CNT, $n = 10$) and $APOE3^{Ch}$ carriers ($APOE3^{Ch}$, $n = 10$), neither of which carry the $PSEN1^{E280A}$ mutation; $PSEN1^{E280A}$-*APOE3* carriers with MCI (MCI $PSEN1^{E280A}$-*APOE3*, $n = 10$); $PSEN1^{E280A}$-*APOE3* carriers without MCI (non-MCI $PSEN1^{E280A}$-*APOE3*, $n = 10$); and $PSEN1^{E280A}$-$APOE3^{Ch}$ carriers without MCI (non-MCI $PSEN1^{E280A}$-$APOE3^{Ch}$, $n = 9$, including one individual homozygous for the Christchurch mutation). Participants were excluded if they were pregnant, had active infectious symptoms, had autoimmune or inflammatory disorders, or reported consumption of psychoactive substances. Groups were matched for sex and age, except for the MCI group, whose participants were required to be older to reflect the typical age of onset for MCI. Experimental procedures were performed under blinded conditions: for $PSEN1^{E280A}$ carriers, the experimenter was unaware of the cognitive status of participants (MCI vs. non-MCI), and for $APOE3^{Ch}$ carriers, the $PSEN1^{E280A}$ genotype was blinded. A schematic overview of the study groups is provided in Figure S1A in supporting information.

The reported *n* refers to individual patients. This information corresponds to the initial sample sizes; however, the number of samples analyzed may vary slightly across experiments due to limited sample availability. The patient IDs corresponding to the samples analyzed in each experiment are indicated in the figure legends (Table S1 in supporting information). Detailed information on sex, age at neuropsychological evaluation, neuropsychological test performance (Mini-Mental State Examination [MMSE], Functional Assessment Staging Test [FAST], and Consortium to Establish a Registry for Alzheimer's Disease [CERAD] word list delayed recall), and comorbidities is available in Table S1. Comorbidities exhibited a mild age-related trend, with no detectable dependence on *APOE* or *PSEN1* genotype. The MMSE is a 30-point screening scale that assesses global cognitive function across domains such as orientation, attention, memory, language, and visuospatial skills.[30] The FAST scale evaluates functional decline in dementia by categorizing individuals into seven stages based on their ability to perform daily activities.[31] The CERAD word list delayed recall assesses episodic memory by testing the ability to recall a list of words after a delay, making it particularly sensitive to early memory impairment.[32] Comparative analyses throughout the study are structured as follows: first, by *APOE* and *PSEN1* genotype; second, by cognitive status within $PSEN1^{E280A}$-*APOE3* carriers; and third, between non-MCI $PSEN1^{E280A}$ carriers with *APOE3* and $APOE3^{Ch}$.

2.2 | Isolation of EVs

EVs were isolated from peripheral blood collected in citrate-containing anticoagulant tubes (BD Vacutainer), following a modified version of a previous protocol,[24] as outlined in the workflow shown in Figure S1A. All procedures were performed at room temperature unless otherwise specified. Briefly, whole blood was centrifuged at 1800 × g for 10 minutes to separate the plasma. Plasma was then centrifuged at 3000 × g for 20 minutes to obtain platelet-poor plasma (PPP). PPP was diluted 1:5 in Dulbecco's phosphate-buffered saline (DPBS, 1X; Gibco BRL), one volume of PPP plus four volumes of DPBS. The mixture was aliquoted into 1 mL fractions and centrifuged at 17,000 × g for 1 hour to enrich EVs. After centrifugation, 900 µL of the supernatant was removed, and the EV pellets in multiple aliquots were immediately stored at −80°C for later characterization or use in cell stimulation assays. No further freeze–thaw cycles were done per aliquot.

2.3 | Scanning electron microscopy

EVs were washed twice with 0.1X DPBS, prepared by diluting DPBS in distilled water at a 1:9 ratio. Each wash was performed by centrifugation at 17,000 × g for 1 hour at room temperature. The EVs were then resuspended in 20 µL of 0.1X DPBS and fixed with glutaraldehyde (G6403, Sigma-Aldrich) at a 3:1 ratio relative to the DPBS volume. After fixation, samples were washed again under the same centrifugation conditions. The final pellet was resuspended in 20 µL of 0.1X DPBS. A drop of the EV suspension was placed on a suitable support, air-dried at room temperature, and prepared for imaging. Samples were analyzed using a JEOL JSM-6490LV scanning electron microscope, operated at 30 kV with a resolution of 3 nm and a maximum magnification of 30,000×. Representative images were obtained to evaluate vesicle morphology, diameter, and structural integrity.

2.4 | Western blot analysis

Western blot analysis was performed on EVs from *APOE3*^{Ch}, *PSEN1*^{E280A} with *APOE3*, *APOE3*^{Ch}, and *APOE3*$^{Ch/Ch}$ patients, along with a plasma sample used as a control. EVs were lysed in radioimmunoprecipitation assay lysis buffer containing a protease and phosphatase inhibitor cocktail (Cytoskeleton, Biochem), and protein concentrations were determined using the Pierce BCA protein assay kit (Thermo Fisher Scientific). For each sample, 50 µg of total protein were loaded. Samples were prepared under denaturing conditions in 1x Laemmli buffer (Boston Bioproducts) and separated by sodium dodecyl sulfate polyacrylamide gel electrophoresis 4% to 20% (Bio-Rad) at 90 V for 1.5 hours. Proteins were transferred onto polyvinylidene fluoride membranes using a dry-transfer system (iBlot, Thermo Fisher). Membranes were blocked for 1 hour with Odyssey blocking buffer (LI-COR) and probed with anti-CD81 (10630D, Invitrogen, Table S2 in supporting information) and anti-CD63 (10628D, Invitrogen, Table S2). After three washes in Tris-buffered saline Tween-20 (ThermoFisher Scientific), membranes were incubated for 1 hour at room temperature with horseradish peroxidase (HRP)-conjugated secondary anti-mouse (R&D Systems, 1:2000, Table S2). After additional washes, protein detection was performed using a SuperSignal West Femto Maximum Sensitivity Substrat (Thermo Fisher) and visualized on an iBright FL1500 (Thermo Fisher).

2.5 | Nanotracking analysis

EVs from four individuals per study group were thawed, resuspended in 1 mL of DPBS, and analyzed using a NanoSight LM10 system (Malvern Instruments) equipped with a 405 nm blue laser and an sCMOS camera. Particle tracking analysis was performed using NanoSight NTA software, version 3.4 Build 3.4.4 (Malvern Panalytical Ltd). For each sample, five videos of 60 seconds each were recorded under standardized acquisition settings: a frame rate of 25 frames per second (fps) and a detection threshold of 3. The software generated particle size distribution profiles and concentration measurements, reporting mean particle size, as well as D10, D50 (median), and D90 values.

2.6 | Flow cytometry

2.6.1 | Total and size-specific EV count

The concentration and size distribution of EVs were also evaluated by flow cytometry. EVs were thawed, resuspended in 300 µL of DPBS, and acquired on a CytoFLEX cytometer (Beckman Coulter) using CytExpert v2.6. Data were collected using the Violet-SSC-H parameter. EV counts (EVs/µL) were corrected by subtracting electronic noise, determined from DPBS-only controls, and adjusted for plasma volume and dilution. Size estimation was performed using Megamix-Plus beads of 0.1, 0.16, 0.20, 0.24, 0.3, and 0.5 µm.

2.6.2 | Cell-type EV source

To evaluate the cellular source, membrane composition, and possible cargo content of systemic EVs, EVs were stained with human-specific antibodies and lipid-intercalating probes, followed by flow cytometry analysis, as previously described.[33,34] As controls, unstained EVs were included to assess basal signal, and phosphate-buffered saline (PBS) plus antibodies was used to monitor potential false-positive events. The following markers were used: anti-CD45 FITC (Clone HI30, BD Biosciences, Table S2), anti-CD105 PE (Clone 43A3, BioLegend, Table S2), anti-CD235a Pacific Blue (Clone HI264, BioLegend, Table S2), anti-CD41a FITC (Clone HIP8, BioLegend, Table S2), anti-aquaporin 4 (AQP4) FITC (Bioss, Table S2), anti-CD171 BV421 (Clone 5G3, BD, Table S2), and anti-CD140b (PDGFRβ) PE-Cy7 (Clone APB5, Invitrogen, Table S2). Phosphatidylserine exposure was assessed using Annexin V Cy5 (BD, Table S2) in Annexin V binding buffer (BD).

Additionally, EV membranes were labeled using FM4-64 (T13320, Invitrogen, Table S2), a dye that intercalates into lipid bilayers, and BODIPY 493/503 (D-3922, Thermo Fisher Scientific, Table S2), which has been shown by flow cytometry to be more specific for detecting cellular lipid droplets.[35] The presence of apoE in the EV-enriched fraction was evaluated using an anti-ApoE PE (Clone E6D7, BioLegend, Table S2). For antibody staining, one aliquot of the EV per patient was used for each of the following four panels: Panel 1—cluster of differentiation (CD)45, CD105, and CD235a; Panel 2—annexin, CD41a, and apoE; Panel 3—AQP4 and platelet-derived growth factor beta (PDGFRβ); and Panel 4—BODIPY and CD171. To each aliquot, 100 µL of a prepared antibody solution in DPBS was added, and samples were incubated for 30 minutes at room temperature in the dark. Samples were washed with 1 mL of DPBS at 17,000 × g for 60 minutes to remove excess antibodies. For FM4-64 staining, 5 µL of a 1 µg/µL solution prepared in molecular-grade water were added, incubated at 37°C for 30 minutes, and washed under the same conditions.

All acquisitions were performed using an LSRFortessa flow cytometer (BD Biosciences) and FACSDiva software. Mean fluorescence intensity (MFI) was calculated within the EV gate and represented as fold change relative to the control group. Flow cytometry data were analyzed using FlowJo 7.6.1 software (Tree Star, Inc.).

We analyzed the cell-type sources of systemic EVs by quantifying the percentage of EVs positive for CD45 (leukocytes), CD105 (endothelial cells), CD235a (erythrocytes) and CD41a (platelets), PDGFRβ (pericytes), AQP4 (astrocytes), and assessed their relative marker levels using MFI.

2.7 | Lipid/lipoprotein assays in plasma and matched EV samples

Lipid and lipoprotein measurements were performed on both plasma and EV preparations. For *APOE3*^{Ch} carriers, lipid analyses were conducted in all available participants. For the non-*APOE3*^{Ch} groups, analyses were performed in a representative subset consisting of four cognitively unimpaired controls and eight *PSEN1*^{E280A}-*APOE3* carriers (four MCI and four non-MCI), matched by age and sex. Total cholesterol, high-density lipoprotein (HDL) cholesterol, triglycerides, low-density lipoprotein (LDL) cholesterol, and very-low-density lipoprotein (VLDL) cholesterol were quantified using standard enzymatic colorimetric clinical assays following manufacturers' instructions. Total cholesterol was measured using a cholesterol esterase–cholesterol oxidase–peroxidase (CHOD-POD) method; HDL cholesterol using a direct HDL cholesterol oxidase assay; and triglycerides using a lipase/glycerol phosphate oxidase–p-aminophenazone (GPO-PAP) method without free-glycerol correction. LDL and VLDL cholesterol concentrations were calculated using the Friedewald equation. EV suspensions obtained from the same participants were assayed in parallel using the same enzymatic kits with appropriate dilutions, allowing direct comparison of lipid and lipoprotein content between plasma and EV fractions.

2.8 | Liquid chromatography tandem mass spectrometry analysis

For proteomic analysis, EV samples were resuspended in 50 to 100 µL of PBS. Each sample was mixed with 40 µL of 8 M urea/50 mM Tris-HCl solution, followed by reduction with 8 mM dithiothreitol (Thermomixer, 3.35 g, 30°C for 60 minutes) and alkylation with 20 mM iodoacetamide (Thermomixer, 3.35 g, 30°C for 30 minutes in the dark). Samples were subsequently diluted with 50 mM Tris-HCl to reduce the urea concentration to < 2 M, a critical step to avoid inhibition of trypsin activity. Digestion was initiated by adding 10 µL of modified sequencing-grade trypsin (20 ng/µL, Promega) prepared in PBS, and samples were incubated overnight at 37°C. After digestion, samples were acidified by adding 10 µL of a 20% formic acid solution and desalted using StageTip-based purification.

On the day of mass spectrometry analysis, samples were reconstituted in 10 µL of high-performance liquid chromatography (HPLC) solvent A (0.1% formic acid in water). Peptides were separated using a custom-packed nanoscale reverse-phase HPLC column (2.6 µm C18 silica beads packed into a fused silica capillary, 100 µm inner diameter, ≈ 30 cm length, with a flame-drawn tip). Sample loading was performed using a Famos autosampler (LC Packings). A linear gradient of increasing solvent B (97.5% acetonitrile, 0.1% formic acid) was applied for peptide elution. Eluted peptides were ionized by electrospray and analyzed with an Orbitrap Exploris 480 mass spectrometer (Thermo Fisher Scientific). Peptides were detected, isolated, and fragmented to generate tandem mass spectra. Peptide sequences and protein identities were assigned by matching acquired spectra against a protein database using the software program Sequest (Thermo Fisher Scientific).[36] Database searches included reversed sequences, and results were filtered to achieve a peptide-level false discovery rate (FDR) between 1% and 2%. The analysis was conducted at the Taplin Biological Mass Spectrometry Facility (Boston, MA).

2.9 | Bioinformatic analysis

2.9.1 | Processing data

The initial processing of proteomic data was performed using the Perseus software platform (version 2.1.3.0). The raw matrix, containing vesicle-associated proteins identified across the study groups, was loaded into Perseus. Prior to further analysis, the intensity values were subjected to a log2 transformation. A row-based filter was then applied to retain only proteins with at least five valid intensity values within each experimental group. After filtering, a final list of 189 proteins was obtained. Missing intensity values (NaN values) were imputed using the MinDet method, replacing missing entries with minimal deterministic values inferred from the data distribution. To map previously reported EV proteins, we used the most recent versions of Vesiclepedia (version 5.1, 2023) and ExoCarta (version 6, 2025).

2.9.2 | Cellular components profile of proteomic data

The analysis of cellular components based on identified proteins was performed using FunRich software (version 3.1.4). The uploaded protein list was compared against the integrated cellular component database within FunRich. Default settings were used for the analysis, and enrichment results were expressed as the percentage of total identified proteins mapped to each cellular category.

2.9.3 | Partial least squares discriminant analysis

Multivariate statistical analyses were conducted using partial least squares discriminant analysis (PLS-DA),[37] a method particularly well suited for datasets with a small number of samples and a large number of variables. We applied PLS-DA to visualize group differences by reducing dimensionality while retaining class-discriminative patterns in the EV proteomic data. The PLS-DA was performed following established procedures.[38] PLS-DA was performed using scikit-learn's PLSRegression module in Python to identify features that discriminate among groups. Three components were extracted, and the resulting component scores were used to visualize group separation in 3D. Group-specific ellipsoids were constructed using mean-centered covariances, and explained variance for each component was calculated. The Fisher Discriminant Index was used to quantify separation between groups. Variable Importance in Projection (VIP) scores were calculated to assess the contribution of each variable to the model. The VIP is a weighted sum of the squared PLS weights, reflecting both the variance explained and the covariance between predictors and the dependent variable (i.e., group classification). For comparison, we also included the significance Multivariate Correlation (sMC) metric, which, unlike VIP, excludes residual variance from the predictors.

2.9.4 | Differential expression proteins and ingenuity pathway analysis

Protein fold-changes were calculated using a threshold of 1.5, meaning that only proteins with expression levels increased or decreased by at least 1.5-fold were considered for subsequent analysis. Ingenuity Pathway Analysis (IPA, QIAGEN) was used to explore the biological relevance of differentially enriched proteins identified in EVs across the groups of interest. Differentially enriched proteins (DEPs) were defined using FDR-adjusted *P* values, and these FDR-significant proteins served as the input for IPA. Proteins with significant fold changes and associated *P* values were uploaded into IPA, which mapped them to its curated knowledge base. The analysis included the identification of enriched canonical pathways, upstream regulators, and molecular interaction networks. Core analysis was performed using default parameters, with both direct and indirect relationships considered. The software computed enrichment scores based on the overlap between the input dataset and known pathway components, and significance was assessed using a right-tailed Fisher exact test. Pathways were ranked according to *P* value and activation *z* score, the latter predicting pathway activation or inhibition based on the directionality of protein changes. Only pathways meeting defined significance thresholds ($P < 0.05$) were included in the analysis.

2.9.5 | Mapping reads to the genome

We used the data set published previously (GSE222494 and GSE222495) and created a pre-mRNA reference using the GRCh38 (version 1.2.0) genome provided by 10X Genomics, following the official reference-building instructions available at 10X Genomics support. Sequencing reads obtained from human samples were aligned to this human pre-mRNA reference using the Cell Ranger ARC pipeline from 10X Genomics (version 2.0.2) with default settings.

2.9.6 | Eliminating background noise counts

To reduce ambient RNA contamination in the single-nucleus RNA sequencing (snRNA-seq) datasets, we applied the CellBender "remove-background" function (version 0.2) from the Broad Institute, which is available at CellBender GitHub. This tool was used to detect non-empty droplets and suppress background signal. The specific parameters used were: –expected-cells 10,000, –total-droplets-included 30,000, –learning_rate 2.5e-6, and –low-count-threshold 100.

2.9.7 | Clustering and cell-type identification in snRNA-seq and scRNA-seq datasets

All statistical and computational analyses were conducted using R (version 4.3.2) or Python (version 3.8). We started by performing filtering and quality control on the gene expression matrices. The filtered count matrices for each individual sample were merged into a unified Scanpy AnnData object (Scanpy version 1.6.0, AnnData version 0.7.4) for downstream processing, using default parameters unless otherwise noted. The filtering criteria applied included: (1) detection and removal of doublets using Scrublet (version 0.2.2), (2) exclusion of nuclei with < 300 detected genes, (3) removal of genes not present in at least 100 nuclei across all samples, (4) removal of nuclei expressing < 300 genes, (5) exclusion of genes with < 5 total counts, (6) removal of nuclei with > 7900 detected genes, (7) removal of nuclei with mitochondrial gene content $> 5\%$, and (8) exclusion of mitochondrial genes and the highly expressed long non-coding RNA MALAT1.

To correct for potential biases due to differences in sequencing depth among samples, we downsampled the reference dataset to 1000 unique molecular identifiers (UMIs) per nucleus, following protocols outlined previously.[39,40] Batch effects were addressed using the Harmony package (version 0.1.0) with default parameters. Next, the

expression matrix was normalized to counts per million (CPM) and then log-transformed using a natural log with a pseudocount of one.

We identified the top 2000 highly variable genes and performed principal component analysis (PCA) to retain components explaining 90% of the variance. Using the Scanpy preprocessing module, we built a neighborhood graph based on 300 nearest neighbors and 26 PCA components. A two-dimensional embedding was created via Uniform Manifold Approximation and Projection (UMAP). Subsequently, clustering was performed using the Leiden algorithm at a resolution of 0.05, applied to the neighborhood graph. This yielded seven distinct clusters at the whole-tissue level.

For cell-type annotation, we used a combination of gene marker mapping from published literature and annotations from external reference databases, including CellMarker 2.0, Tabula Sapiens, PanglaoDB, and expanded Azimuth datasets. After cell types were assigned, we carried out differential gene expression (DGE) analysis for each cell type. We applied a pseudo-bulk approach using the edgeR package, which fits generalized linear models (GLMs) and performs likelihood ratio tests to determine differential expression. Fold changes were calculated by comparing each sample's normalized gene expression counts to the average of the control group. Genes were classified as differentially expressed when the adjusted P value was < 0.05.

2.10 | Cell cultures

2.10.1 | Endothelial cells

Primary human brain microvascular endothelial cells (HBMECs; ACBRI 376, Cell Systems) were used as an in vitro model of human endothelial cells. Cells were thawed and maintained in Endothelial Cell Growth Media (CCM027, R&D Systems) with Normocin (InvivoGen) at 37°C in a humidified atmosphere containing 5% CO_2. For subculturing, cells were detached using Detachin Cell Detachment Reagent (T100106, Fisher Scientific) and seeded onto 12 mm gelatin-coated coverslips at a density of 2.5×10^5 cells per coverslip.

2.10.2 | Astrocytes

Human astrocytes (SVG p12 cell line; CRL-8621, ATCC) were used as an in vitro astrocyte model. Cells were maintained in Dulbecco's Modified Eagle Medium (DMEM) supplemented with GlutaMAX (Thermo Fisher Scientific), 10% fetal bovine serum (FBS; ATCC), and Normocin. Cultures were incubated at 37°C in a humidified atmosphere containing 5% CO_2 and were routinely passaged upon reaching 80% to 90% confluency. For subculturing, cells were detached using Detachin and seeded onto 12 mm gelatin-coated coverslips at a density of 2.5×10^5 cells per coverslip.

2.10.3 | HBMECs and astrocyte cocultures

Human astrocytes were seeded onto gelatin-coated coverslips containing paraffin spots (≈ 0.5 mm in height and 2 mm in width), while HBMECs were seeded onto separate gelatin-coated coverslips without paraffin spots. On day in vitro 3 (DIV 3), the astrocyte coverslips and the endothelial cells coverslips were placed together into the same well of a 12-well plate, with the cell-coated sides facing each other. Although the two cell types shared the same culture medium, the paraffin spots physically prevented direct cell–cell contact. Cocultures were maintained for five additional days. On DIV 8 for both cell types, cocultures were stimulated for 24 hours with EVs ($n = 4$; representing EV samples isolated from four biologically independent donors per group) or recombinant ApoE4 (1 μM), ApoE3 and ApoE3Ch (1 μM) proteins (INNOVAGEN) resuspended in EBM-2 Endothelial Cell Growth Basal Medium-2 (CC-3156, Lonza), previously filtered through 0.1 μm Sterile Disposable Filter Units (565-0010, Thermo Scientific). EVs were added at an approximate ratio of one vesicle per target cell.

2.11 | Transendothelial electrical resistance measurements

HBMECs at passage 5 were seeded onto 0.4 μm pore size, gelatin-coated Transwells in a 12-well plate format (3460, Costar) at a density of 7.5×10^4 cells/well. Cells were maintained in microvascular endothelial growth medium-2 (EGM-2 MV, CC-3202, Lonza) containing EBM-2 Basal Medium (CC-3156, Lonza) supplemented with EGM-2 MV SingleQuots (CC-4147, Lonza). Cultures were incubated at 37°C in a humidified atmosphere containing 5% CO_2. After 2 hours, lithium chloride (LiCl, L9650, Sigma Aldrich) was added to the medium to reach a final concentration of 10 mM. The next day, the medium was replaced with fresh EGM-2 containing 10 mM LiCl and 550 nM hydrocortisone (H0888, Sigma Aldrich), and cells were incubated for an additional 24 hours.

Cells were then stimulated with 50 ng/mL vascular endothelial growth factor (VEGF, 293-VE, R&D Systems) in combination with either 1 μM wild-type APOE3 or 1 μM APOE3Ch, diluted in experimental medium consisting of EBM-2 supplemented with 5% FBS (30-2020, ATCC), 1% GlutaMAX (35050061, Thermo Fisher Scientific), 0.2% Normocin, and 10 mM LiCl. Vehicle controls were prepared by adding DPBS at the equivalent volume of APOE3 to the experimental medium. Transendothelial electrical resistance (TEER) was monitored for 48 hours using a cellZscope+ (N cZs+, NanoAnalytics). TEER values ($\Omega \cdot cm^2$) were calculated according to the manufacturer's instructions by multiplying the raw resistance by the surface area of the monolayer, and background TEER, measured from wells containing only experimental medium, was subtracted from all readings.

2.12 | Lactate dehydrogenase release cytotoxicity assay

To test cytotoxicity changes in response to EVs from the different study groups, cytotoxicity was assessed by measuring lactate dehydrogenase (LDH) release using the Roche Cytotoxicity Detection Kit (Cat. #4744926001, Millipore). On day six of coculture (DIV 9 for both astrocytes and endothelial cells), after 24 hours of EV treatment, culture supernatants were collected, and LDH activity was quantified based on the linear rate of NADH consumption during the enzymatic conversion of pyruvate to lactate, measured spectrophotometrically. The percentage of cytotoxicity for each condition was calculated using the following formula: %Cytotoxicity = ([A – low control] / [high control – low control]) × 100, where A corresponds to the mean LDH activity from three technical replicates per treatment, low control represents spontaneously released LDH from untreated cells, and high control corresponds to maximal LDH release induced by 1% Triton X-100 treatment for 24 hours.

2.13 | Immunofluorescence

For morphological and phenotypical analyses, astrocyte–endothelium cocultures were fixed and stained. Cells were initially fixed in 4% paraformaldehyde for 20 minutes at room temperature. To reduce autofluorescence, samples were incubated in 50 mM ammonium chloride for 10 minutes. Cells were then permeabilized with PBS containing 0.2% Triton X-100 for 5 minutes and blocked with PBS supplemented with 2.5% FBS for 1 hour to prevent non-specific protein interactions.

Primary antibody incubation was performed overnight at 4°C using the following antibodies: anti-β-catenin (sc-7963, Santa Cruz Biotechnology, Table S2), anti-GFAP (ab7260, Abcam, Table S2), and anti-Apolipoprotein E (178479, Invitrogen, Table S2). The next day, cells were incubated with Alexa Fluor 488- or Alexa Fluor 594-conjugated secondary antibodies (Invitrogen, Table S2) for 1 hour at room temperature. Nuclei were counterstained with DAPI (1:5000, Invitrogen) for 1 hour. After final washes with PBS, samples were mounted with FluorSave mounting medium (Millipore) and imaged using a confocal microscope.

2.14 | Confocal imaging

High-resolution images were acquired using a confocal laser scanning microscope (Leica TCS SP8 Microsystems) equipped with a 40× dry objective lens (NA 1.25), at a resolution of 1024 × 1024 pixels. For each cell type, four images were captured per coverslip, with at least three coverslips analyzed per experimental condition. Image deconvolution was subsequently performed using the classic maximum likelihood estimation (CMLE) with a signal-to-noise ratio of 12 in Huygens Professional software, version 19.10 (Scientific Volume Imaging).

2.15 | Morphometric analysis

To quantify gaps between endothelial cells, images were analyzed using ImageJ FIJI v2.16.0. software. β-catenin-stained images were binarized by applying a consistent threshold across all experimental conditions. The resulting masked regions, corresponding to areas lacking β-catenin signal, were considered gaps. Particle size analysis was performed within the masked regions to determine both the number and size of individual gaps (four fields per coverslip per condition). To assess nuclear β-catenin localization, DAPI and β-catenin immunofluorescence signals were segmented separately using intensity thresholding based on the Yen algorithm in FIJI. Total stained areas and colocalizing regions were measured. Colocalized signals between DAPI and β-catenin were identified using the "AND" function from the Image Calculator tool in FIJI. The mean intensity of glial fibrillary acidic protein (GFAP) fluorescence was quantified in GFAP-positive images (40×) using the FIJI "Count and measure objects" tool. To quantify ApoE-positive particles, images were segmented using the Yen thresholding algorithm in FIJI. Particle detection and quantification were performed using the "Analyze Particles" tool, applying consistent settings across all images. To normalize the number of ApoE-positive particles, the particle count in each field was divided by the number of nuclei (DAPI-stained), allowing the expression of results as the number of ApoE particles per cell.

2.16 | Statistical analysis

The sample size (n) for each experimental group corresponded to an individual patient, and each patient was treated as an independent experimental unit. Data distribution was assessed for normality using the Shapiro–Wilk test. For comparisons between two groups, either an unpaired t test or a Mann–Whitney U test was applied, depending on data distribution. For comparisons of more than two groups, parametric univariable data were analyzed using one-way analysis of variance followed by Tukey post hoc multiple comparisons test. Nonparametric data were evaluated using the Kruskal–Wallis test followed by Dunn multiple comparisons test. For the correlation analysis, we used Python version 3.12.7 and implemented the pandas (v2.2.3) and scipy (v1.13.1) libraries. Data preprocessing and transformation were performed using pandas according to the specific requirements of our analyses. To compute the Pearson correlation coefficients and corresponding P values for testing the null hypothesis of no correlation, we used the scipy.stats.pearsonr function. This function calculates the Pearson coefficient by assessing the linear relationship between two datasets and returns the P value for the hypothesis test that the underlying distributions of the samples are uncorrelated and normally distributed. We used the seaborn library (v0.13.2), specifically the seaborn.regplot function, to generate scatter plots, fit models, and display the resulting lines along with 95% confidence intervals.

The specific statistical tests used for each figure are indicated in the respective figure legends. Results are presented using boxplots, violin

plots, heatmaps, and scatter plots. The statistical analyses, including normality testing and group comparisons, were performed using GraphPad Prism version 9.0.0 (Dotmatics). Comparisons against the control group (CNT) are indicated with a number sign (#), and comparisons against the non-MCI *PSEN1*^{E280A}–*APOE3* group are indicated with an asterisk (*). Statistical significance was denoted as */# $P < 0.05$, **/## $P < 0.01$, and ***/### $P < 0.001$.

3 | RESULTS

3.1 | *APOE3*^{Ch}-associated reduction in endothelial and leukocyte markers in EVs

In accordance with Minimal Information for Studies of Extracellular Vesicles 2023 guidelines, we verified that the isolated material contains both small and large EVs, with similar size and concentration across all groups (Figure S1).[41]

The *PSEN1*^{E280A}-*APOE3*^{Ch} carriers showed lower percentages of CD105+, CD235a+, and CD45+CD105+ EVs compared to *PSEN1*^{E280A}-*APOE3*. The reduction in CD235a+ and CD45+CD105+ EVs was also observed in *APOE3*^{Ch} carriers without the *PSEN1*^{E280A} mutation, suggesting that this effect may be associated with *APOE3*^{Ch} independently of the *PSEN1* variant (Figure 1A). These suggest that *APOE3*^{Ch} may influence the release or stability of endothelial (CD105), erythrocyte (CD235a)-, and leukocyte-endothelial (CD45+CD105+)-derived EVs. With respect to marker levels, EVs from *APOE3*^{Ch} carriers showed reduced levels of CD45 and CD105 compared to controls, independently of the *PSEN1* variant while CD41a levels were increased (Figure 1B), suggesting a potential shift in platelet-associated EV profiles.

We next stratified *PSEN1*^{E280A}-*APOE3* carriers by cognitive status (MCI vs. non-MCI) to identify potential early alterations associated with clinical impairment. Non-MCI *PSEN1*^{E280A}-*APOE3* carriers showed high interindividual variability in CD45+ and CD105+ EVs (Figure 1C). Their EVs also exhibited higher CD45, CD105, and CD235a levels than those from individuals with MCI (Figure 1D). These patterns may reflect subtle molecular alterations present in these cell types prior to the onset of clinical impairment.

We compared EVs from non-MCI *PSEN1*^{E280A}-*APOE3* to those from non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch} carriers to evaluate whether the *APOE3*^{Ch} variant modulates systemic EV phenotypes in the preclinical ADAD context. This analysis revealed a consistent pattern with the results from comparisons across genotypic groups, showing reduced percentages of CD105+, CD235a+, and CD45+CD105+ EVs in *APOE3*^{Ch} carriers (Figure 1E). Notably, the patient homozygous for *APOE3*^{Ch} with *PSEN1*^{E280A} consistently exhibited some of the lowest values within the group. EVs from *APOE3*^{Ch} carriers also showed lower levels of CD45, CD105, and CD235a, and higher of CD41a (Figure 1F).

These findings suggest that *PSEN1*^{E280A}-*APOE3* EVs exhibit elevated levels of endothelial and leukocytes markers before the MCI, while *PSEN1*^{E280A}-*APOE3*^{Ch} carriers show reduced levels of these markers but increased abundance of a platelet-associated marker, indicating a shift in the profile of systemic EVs.

3.2 | Reduced astrocytic and pericytes EV signatures in EVs from *APOE3*^{Ch} carriers

We examined NVU-associated cell type markers in the EVs PDGFRβ, AQP4, and CD171 to determine whether *APOE3*^{Ch} influences the most common circulating EVs released by cells from the NVU.

The percentage of CD171+ EVs was increased in *APOE3*^{Ch} carriers compared to controls, and likewise, in *PSEN1*^{E280A}-*APOE3*^{Ch} compared to *PSEN1*^{E280A}-*APOE3* (Figure 2A). Also, we observed that EVs from *APOE3*^{Ch} carriers showed lower PDGFRβ levels than controls, and similarly, exhibited reduced AQP4 and CD171 levels, regardless of *PSEN1* condition (Figure 2B).

When stratifying *PSEN1*^{E280A}-*APOE3* carriers by cognitive status, the heatmap of EV marker percentages revealed high interindividual variability, preventing the identification of a clear overall pattern (Figure 2C). However, non-MCI *PSEN1*^{E280A}-*APOE3*-EVs exhibit a higher abundance of AQP4 compared to the MCI group (Figure 2D).

We compared EVs from non-MCI *PSEN1*^{E280A}-*APOE3* and -*APOE3*^{Ch} carriers to evaluate the effect of *APOE3*^{Ch} on NVU-derived EVs prior to MCI. Consistent with genotypic comparisons, PDGFRβ+ and AQP4+ EVs were similar between groups, whereas CD171+ EVs were increased in *APOE3*^{Ch} carriers (Figure 2E). Notably, we found that PDGFRβ, AQP4, and CD171 levels were lower in non-MCI *PSEN1*^{E280A}- *APOE3*^{Ch} EVs compared to non-MCI *PSEN1*^{E280A}-*APOE3* EVs (Figure 2F).

Overall, *APOE3*^{Ch} is associated with increased CD171+ EVs and reduced astrocytic and pericytes cell markers in peripheral EVs, even in *PSEN1*^{E280A} carriers. In contrast, non-MCI *PSEN1*^{E280A} -*APOE3* carriers show a higher abundance of AQP4 in EVs than those with MCI. These NVU-EV changes could reflect early *APOE*- and cognition-related shifts in neurovascular coupling.

3.3 | *APOE3*^{Ch} variant is associated with increased apoE levels and neutral lipids in EVs

We assessed neutral lipids by BODIPY staining and apoE levels in the enriched EV fractions, as *APOE3*^{Ch} may influence EV lipid content.

The percentage of BODIPY+ EVs was increased in *APOE3*^{Ch} carriers, independently of *PSEN1* genotype. A similar increase was observed in CD171+ BODIPY+ double-positive EVs, pointing to an enrichment of CD171-associated, neutral lipid-containing EVs in *APOE3*^{Ch} carriers (Figure 3A). Also, we observed that EVs from *APOE3*^{Ch} carriers exhibited higher apoE abundance compared to controls (Figure 3B). Although this difference is not clearly observed in the Western blot due to the inclusion of only one representative patient per group (Figure 3F), it was consistently detected across multiple samples by flow cytometry.

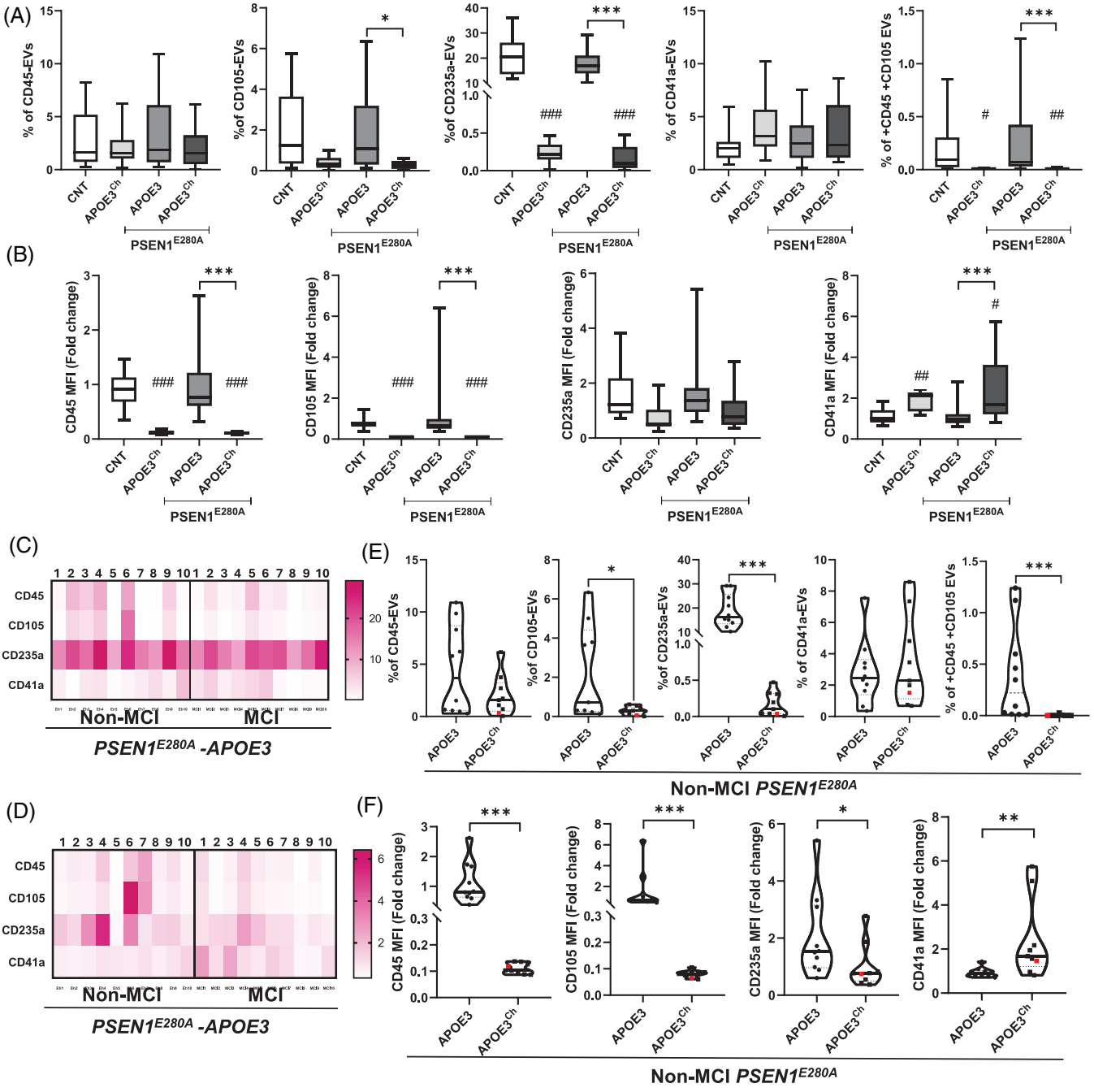


FIGURE 1 Reduction of endothelial and leukocyte EV markers with concomitant platelet marker elevation in *APOE3*^{Ch} carriers. A, Percentage of EVs positive for CD45 (leukocytes), CD105 (endothelial cells), CD235a (erythrocytes), CD41a (platelets), and double-positive CD45-CD105. B, Normalized MFI of EVs positive for markers in (A). C, Heatmap showing EV positivity for the marker panel in (A), comparing *PSEN1*^{E280A}*-APOE3* carriers with and without MCI. D, Heatmap of MFI for *PSEN1*^{E280A}*-APOE3* carriers with and without MCI. E, Violin plots comparing positivity of markers in non-MCI *PSEN1*^{E280A} carriers to *APOE3* and *APOE3*^{Ch}. F, Violin plots comparing the MFI in non-MCI *PSEN1*^{E280A} carriers with *APOE3* and *APOE3*^{Ch}. In (C) and (D), each column in the heatmap is labeled with a number, in which each number represents an individual patient. Representative data from CNT, $n = 10$ (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10); *APOE3*^{Ch}, $n = 10$ (Ch1, Ch2, Ch3, Ch4, Ch5, Ch6, Ch7, Ch8, Ch9, Ch10); MCI *PSEN1*^{E280A}*-APOE3*, $n = 10$ (MCI1, MCI2, MCI3, MCI4, MCI5, MCI6, MCI7, MCI8, MCI9, MCI10); non-MCI *PSEN1*^{E280A} *-APOE3*, $n = 10$ (non-MCI-1, non-MCI-2, non-MCI-3, non-MCI-4, non-MCI-5, non-MCI-6, non-MCI-7, non-MCI-8, non-MCI-9, non-MCI-10) and non-MCI *PSEN1*^{E280A}*-APOE3*^{Ch}, $n = 9$ (E-Ch1, E-Ch2, E-Ch3, E-Ch4, E-Ch5, E-Ch6, E-Ch7, E-Ch8, E-Ch9). Data are presented as boxplots or violin plots. Boxplots display the median, interquartile range, and whiskers representing minimum and maximum values. Violin plots illustrate the full data distribution, with individual data points overlaid and a line indicating the median. The data for the homozygous *APOE3*^{Ch} patient is indicated by a red dot. One-way analysis of variance, Tukey multiple comparison test; unpaired *t* test, Mann–Whitney test were performed. Comparison to CNT is denoted by number sign (#); with non-MCI *PSEN1*^{E280A} *-APOE3* by asterisk (*). */# indicates $P < 0.05$, **/## $P < 0.01$, and ***/### $P < 0.001$. *APOE*, apolipoprotein E; CD, cluster of differentiation; CNT, cognitively unimpaired control; EV, extracellular vesicle; MCI, mild cognitive impairment; MFI, mean fluorescence intensity; PSEN, presenilin

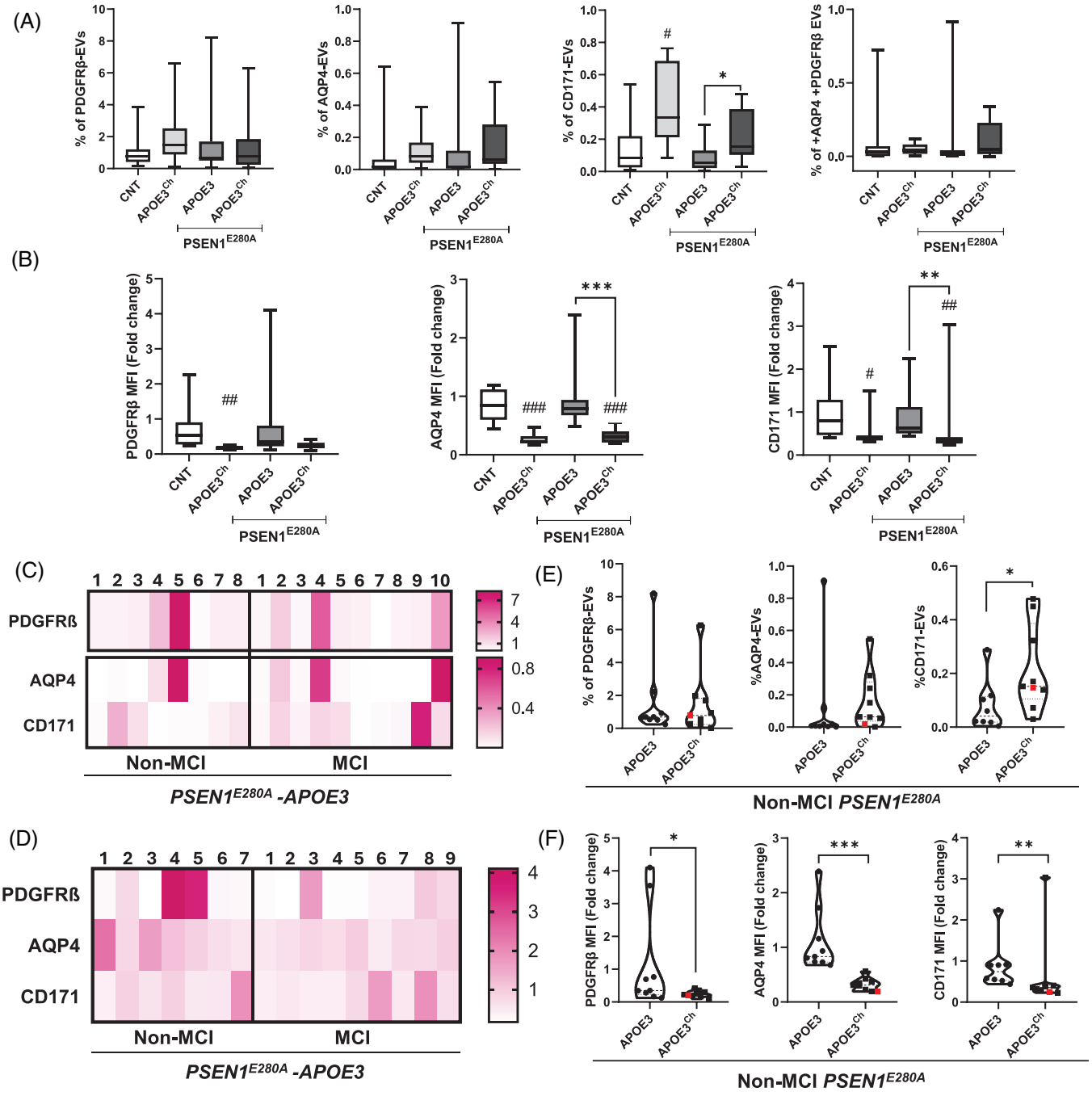


FIGURE 2 *APOE3*^{Ch} is associated with reduced pericyte and astrocyte EV markers alongside increased CD171+ EVs. A, Percentage of EVs positive for PDGFRβ (pericytes), AQP4 (astrocytes), CD171, and double-positive PDGFRβ-AQP4. B, Normalized mean fluorescence intensity (MFI) of EVs positive for markers in (A). C, Heatmap showing EVs positivity for the marker panel in (A), comparing *PSEN1*^{E280A}*-APOE3* carriers with and without MCI. D, Heatmap of MFI for *PSEN1*^{E280A}*-APOE3* carriers with and without MCI. E, Violin plots comparing positivity of markers in non-MCI *PSEN1*^{E280A} carriers to *APOE3* and *APOE3*^{Ch}. F, Violin plots comparing the MFI in non-MCI *PSEN1*^{E280A} carriers to *APOE3* and *APOE3Ch*. In (C) and (D), each column in the heatmap is labeled with a number, where each number represents an individual patient. Representative data from CNT, $n = 10$ (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10); *APOE3*^{Ch}, $n = 10$ (Ch1, Ch2, Ch3, Ch4, Ch5, Ch6, Ch7, Ch8, Ch9, Ch10); MCI *PSEN1*^{E280A}*-APOE3*, $n = 9$ (MCI1, MCI2, MCI3, MCI4, MCI5, MCI6, MCI7, MCI9, MCI10); non-MCI *PSEN1*^{E280A} *-APOE3*, $n = 8$ (non-MCI-1, non-MCI-2, non-MCI-3, non-MCI-4, non-MCI-5, non-MCI-7, non-MCI-8, non-MCI-9) and non-MCI *PSEN1*^{E280A}*-APOE3*^{Ch}, $n = 9$ (E-Ch1, E-Ch2, E-Ch3, E-Ch4, E-Ch5, E-Ch6, E-Ch7, E-Ch8, E-Ch9). Data are presented as boxplots or violin plots. Boxplots display the median, interquartile range, and whiskers representing minimum and maximum values. Violin plots illustrate the full data distribution, with individual data points overlaid and a line indicating the median. The data for the homozygous *APOE3*^{Ch} patient is indicated by a red dot. One-way analysis of variance, Tukey multiple comparison test; unpaired *t* test, Mann–Whitney test were performed. Comparison to CNT is denoted by number sign (#); with non-MCI *PSEN1*^{E280A} *-APOE3* by asterisk (*). */# indicates $P < 0.05$, **/## $P < 0.01$, and ***/### $P < 0.001$. *APOE*, apolipoprotein E; AQP4, aquaporin 4; CD, cluster of differentiation; CNT, cognitively unimpaired control; EV, extracellular vesicle; MCI, mild cognitive impairment; MFI, mean fluorescence intensity; PDGFRβ, platelet-derived growth factor beta; PSEN, presenilin

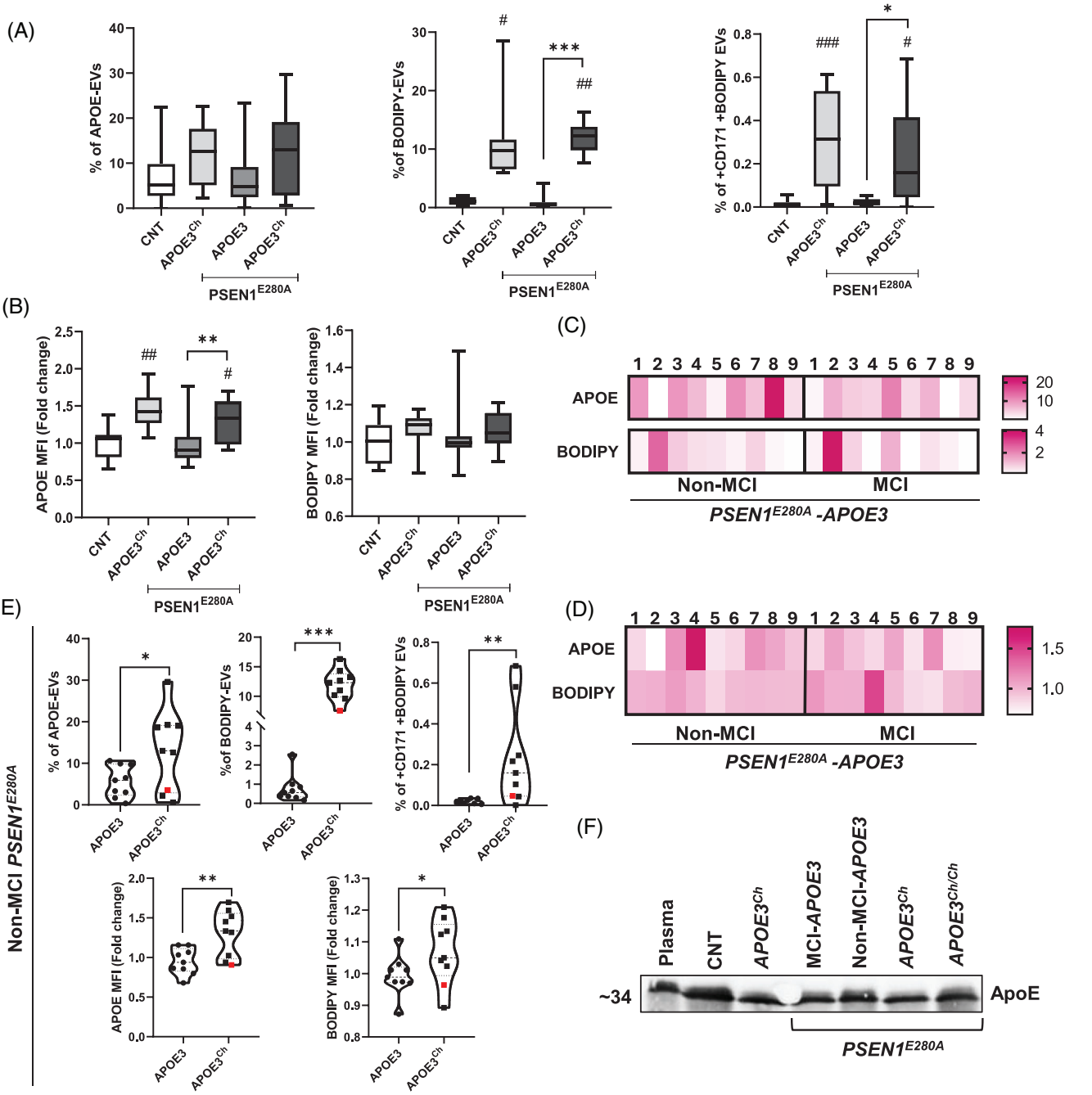


FIGURE 3 ApoE and neutral lipids increase in systemic APOE3^{Ch}-EVs. A, Percentage of positive EVs for *APOE* (apolipoprotein E protein), BODIPY (lipid droplets), and double-positive BODIPY-CD171. B, Normalized mean fluorescence intensity (MFI) of EVs positive for markers in (A). C, Heatmap showing EVs positivity for the marker panel in (A), comparing *PSEN1*^{E280A}*-APOE3* carriers with and without MCI. D, Heatmap of MFI for *PSEN1*^{E280A}*-APOE3* carriers with and without MCI. E, Violin plots comparing positivity of markers (top) and MFI (bottom) in non-MCI *PSEN1*^{E280A} carriers with *APOE3* and *APOE3*^{Ch}. F, Western blot showing apoE levels from a representative patient in each experimental group. In (C) and (D), each column in the heatmap is labeled with a number, where each number represents an individual patient. Representative data from CNT, $n = 10$ (C1, C2, C3, C4, C5, C6, C7, C8, C9, C10); *APOE3*^{Ch}, $n = 10$ (Ch1, Ch2, Ch3, Ch4, Ch5, Ch6, Ch7, Ch8, Ch9, Ch10); MCI *PSEN1*^{E280A}*-APOE3*, $n = 9$ (MCI1, MCI2, MCI3, MCI4, MCI5, MCI6, MCI7, MCI9, MCI10); non-MCI *PSEN1*^{E280A} *-APOE3*, $n = 9$ (non-MCI-1, non-MCI-2, non-MCI-3, non-MCI-4, non-MCI-5, non-MCI-7, non-MCI-8, non-MCI-9, non-MCI-10) and non-MCI *PSEN1*^{E280A}*-APOE3*^{Ch}, $n = 9$ (E-Ch1, E-Ch2, E-Ch3, E-Ch4, E-Ch5, E-Ch6, E-Ch7, E-Ch8, E-Ch9). Data are presented as boxplots or violin plots. Boxplots display the median, interquartile range, and whiskers representing minimum and maximum values. Violin plots illustrate the full data distribution, with individual data points overlaid and a line indicating the median. The data for the homozygous *APOE3*^{Ch} patient is indicated by a red dot. One-way analysis of variance, Tukey multiple comparison test; unpaired *t* test, Mann–Whitney test were performed. Comparison to CNT is denoted by number sign (#); with non-MCI *PSEN1*^{E280A} *-APOE3* by asterisk (*). */# indicates $P < 0.05$, **/## $P < 0.01$, and ***/### $P < 0.001$. *APOE*, apolipoprotein E; CD, cluster of differentiation; CNT, cognitively unimpaired control; EV, extracellular vesicle; MCI, mild cognitive impairment; MFI, mean fluorescence intensity; PSEN, presenilin

PSEN1^{E280A}-*APOE3* individuals stratified by cognitive status revealed that EV positivity and levels of apoE and BODIPY were similar between the MCI and non-MCI groups (Figure 3C, D). We compared non-MCI *PSEN1*^{E280A} carriers to *APOE3* and *APOE3*^{Ch} to evaluate the effect of *APOE3*^{Ch} on neutral lipids and apoE prior to MCI. The percentages of apoE+, BODIPY+, and CD171+/BODIPY+ double-positive EVs and the apoE and BODIPY abundances were increased in *PSEN1*^{E280A}-*APOE3*^{Ch} carriers compared to non-MCI *PSEN1*^{E280A}-*APOE3* carriers (Figure 3E). EVs from the individual homozygous for the *APOE3*^{Ch} variant consistently exhibited among the lowest values across these markers compared to heterozygotes.

To determine whether the increased neutral lipid and apoE content in *APOE3*^{Ch}-EVs reflected systemic lipid alterations, we quantified HDL, LDL, VLDL, triglycerides, and cholesterol in plasma and EVs (Figure S2 in supporting information). Plasma analyses showed higher LDL, VLDL, total cholesterol, and triglycerides in non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch} carriers (Figure S2A), consistent with previously reported mild dyslipidemia in *PSEN1*^{E280A}-*APOE3*^{Ch}.[3,5] However, all lipid species were markedly more abundant in plasma than in EVs (Figure S2B–F), and EV lipoprotein composition did not differ between the two non-MCI *PSEN1*^{E280A} groups (Figure S2G). These findings indicate that plasma lipid differences do not account for the increased neutral lipid and apoE enrichment observed in non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch} EVs.

In summary, *APOE3*^{Ch} is linked to increased CD171+ EVs enriched in neutral lipids and apoE, independent of *PSEN1*^{E280A} status.

3.4 | *APOE3*^{Ch} mitigates the association between peripheral vascular-associated EVs and early cognitive decline in *PSEN1*^{E280A} carriers

Given the distinct EV profile of *PSEN1*^{E280A}-*APOE3*^{Ch} carriers and their delayed MCI onset, we investigated whether EV markers were associated with cognitive performance, potentially reflecting a protective signature. We correlated both the percentage and levels of each marker evaluated in flow cytometry with MMSE, FAST, and CERAD scores.

MMSE scores showed negative correlations with both CD105 and PDGFRβ levels in EVs, as well as with the percentage of CD45+CD105+ EVs, indicating that higher peripheral release of these markers associates with poorer cognitive performance (Figure 4A,B). Conversely, FAST scores exhibited positive correlations with CD45 and PDGFRβ abundance, and the percentages of CD105+ EVs and CD45+CD105+ EVs, linking elevated marker abundance to functional decline (Figure 4A,C). Consistently, CERAD scores inversely correlated with PDGFRβ levels (Figure 4A,D). It is important to acknowledge that several of the observed correlations were influenced by a single outlier sample. When this sample is excluded from the analysis, the only cognition-related association that persists is between MMSE and MFI-CD105 (Table S3 in supporting information).

PDGFRβ levels exhibited consistent positive correlations with vascular and leukocyte markers, suggesting coordinated upregulation of vascular dysfunction and inflammation (Figure S3A,B in supporting information). Conversely, these markers showed inverse correlations with %BODIPY, implying reduced neutral-lipid content in EVs alongside heightened vascular-inflammatory activity (Figure S3A,C). Importantly, the correlations between %BODIPY and MFI CD45, MFI CD105, and MFI AQP4 remain significant after exclusion of the outlier sample (Table S3), suggesting that these associations are more robust and less sensitive to individual variability.

These findings suggest that vascular and leukocyte-derived EV markers are linked to early cognitive decline and may reflect vascular activation, particularly in the absence of *APOE3*^{Ch}.

3.5 | *APOE3*^{Ch}-EVs attenuate vascular, inflammatory, and coagulation-associated cargo enrichment in non-MCI *PSEN1*^{E280A} carriers

We compare EV cargo profiles using liquid chromatography tandem mass spectrometry and identified 189 proteins common to all groups (Figure S4A, Table S4 in supporting information), along with additional group-specific proteins (Table S5 in supporting information). Of these 189 proteins, 127 were previously reported in both the Vesiclepedia and ExoCarta databases, 16 were unique to Vesiclepedia, 7 were exclusive to ExoCarta, and the remaining proteins were not listed in either database (Figure S4B). Cellular component analysis revealed that the common proteins were predominantly extracellular (95%), with 58.7% associated with exosomes, 21% with the plasma membrane, and 14% with lipoprotein particles (Figure S4C). These findings confirm the extracellular and vesicular origin of the protein cargo, consistent with a profile enriched in EVs. The PLS-DA analysis revealed distinct clustering among the study groups, with component 1 accounting for 62.2% of the variance. *APOE3*^{Ch}-EVs formed clusters near each other, while MCI *PSEN1*^{E280A}-*APOE3* clustered closer to CNT. In contrast, non-MCI *PSEN1*^{E280A}-*APOE3* carriers were clearly separated from all other groups, suggesting a distinct EV proteomic signature (Figure S4D).

We then applied PLS-DA to visualize group separation between non-MCI *PSEN1*^{E280A} carriers and CNT individuals, accompanied by the corresponding loading weights (Figure 5A). This was followed by a focused analysis of *PSEN1*^{E280A}-*APOE3* carriers with and without MCI (Figure 5B), and finally, a comparison between non-MCI *PSEN1*^{E280A} carriers with *APOE3* and *APOE3*^{Ch} (Figure 5C). The top five discriminating proteins for each comparison are listed below, with full protein lists and discriminant scores available in Table S6 in supporting information: *PSEN1*^{E280A} carriers versus CNT (amyloid P component, serum [APCS], immunoglobulin lambda constant [IGLC]2, immunoglobulin heavy variable [IGHV]1-2, IGLC1, alpha-1-B glycoprotein [A1BG]); *APOE3* MCI versus non-MCI (apolipoprotein B [APOB], carnosine dipeptidase [CNDP]1, lipopolysaccharide binding protein [LBP], IGLC1, serpin family D member 1 [SERPIND1]), and non-MCI-*PSEN1*^{E280A} groups: *APOE3* versus *APOE3*^{Ch} (CNDP1, APOB, IGLC1, albumin, immunoglobulin heavy variable [IGHV]3OR16-9).

To identify DEPs, we computed fold changes using both FDR-adjusted and unadjusted *P* values, given the small sample size and the number of proteins analyzed. As expected, FDR correction reduced

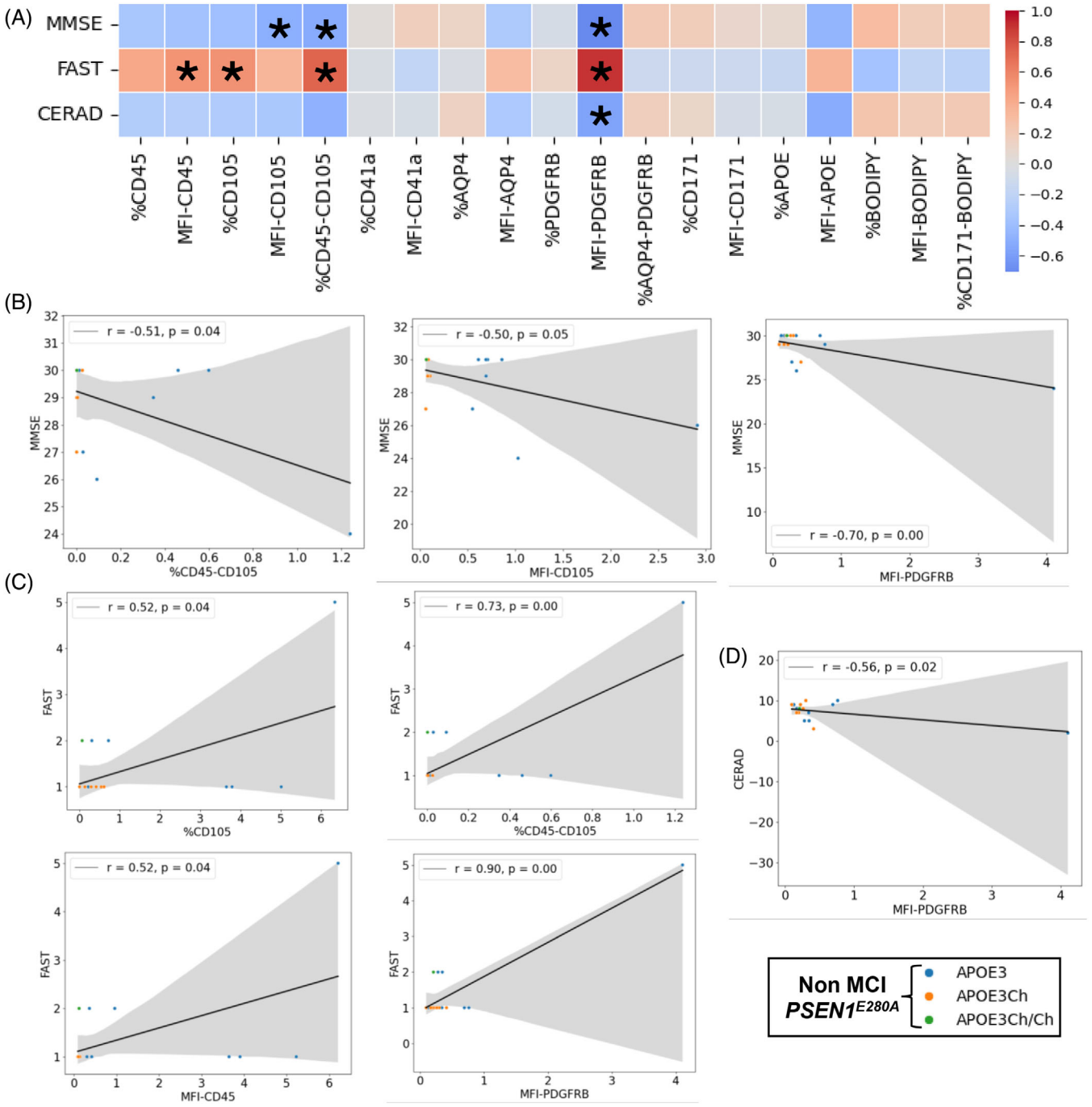


FIGURE 4 *APOE3*^{Ch} mitigates vascular EV–cognition association in *PSEN1*^{E280A} carriers. A, Heatmap showing Pearson correlation coefficients between EV markers (measured by flow cytometry) and neuropsychological test scores: Mini-Mental State Examination (MMSE), Functional Assessment Staging Test (FAST), and Consortium to Establish a Registry for Alzheimer's Disease (CERAD) word list delayed recall score. Asterisks (*) indicate statistically significant correlations ($P \leq 0.05$). Note that higher MMSE and CERAD scores indicate better cognitive performance, whereas higher FAST scores reflect greater functional impairment. Scatter plots illustrate significant correlations between EVs markers with (B) MMSE scores, (C) FAST scores, and (D) CERAD recall scores. Representative data from non-MCI *PSEN1*^{E280A}-*APOE3*, $n = 9$ (non-MCI-1, non-MCI-2, non-MCI-3, non-MCI-4, non-MCI-5, non-MCI-7, non-MCI-8, nNon-MCI-9, non-MCI-10) and non-MCI *PSEN1*^{E280A} -*APOE3*^{Ch}, $n = 9$ (E-Ch1, E-Ch2, E-Ch3, E-Ch4, E-Ch5, E-Ch6, E-Ch7, E-Ch8, E-Ch9). *APOE*, apolipoprotein E; CD, cluster of differentiation; EV, extracellular vesicle; MCI, mild cognitive impairment; MFI, mean fluorescence intensity; PDGFRβ, platelet-derived growth factor beta; PSEN, presenilin

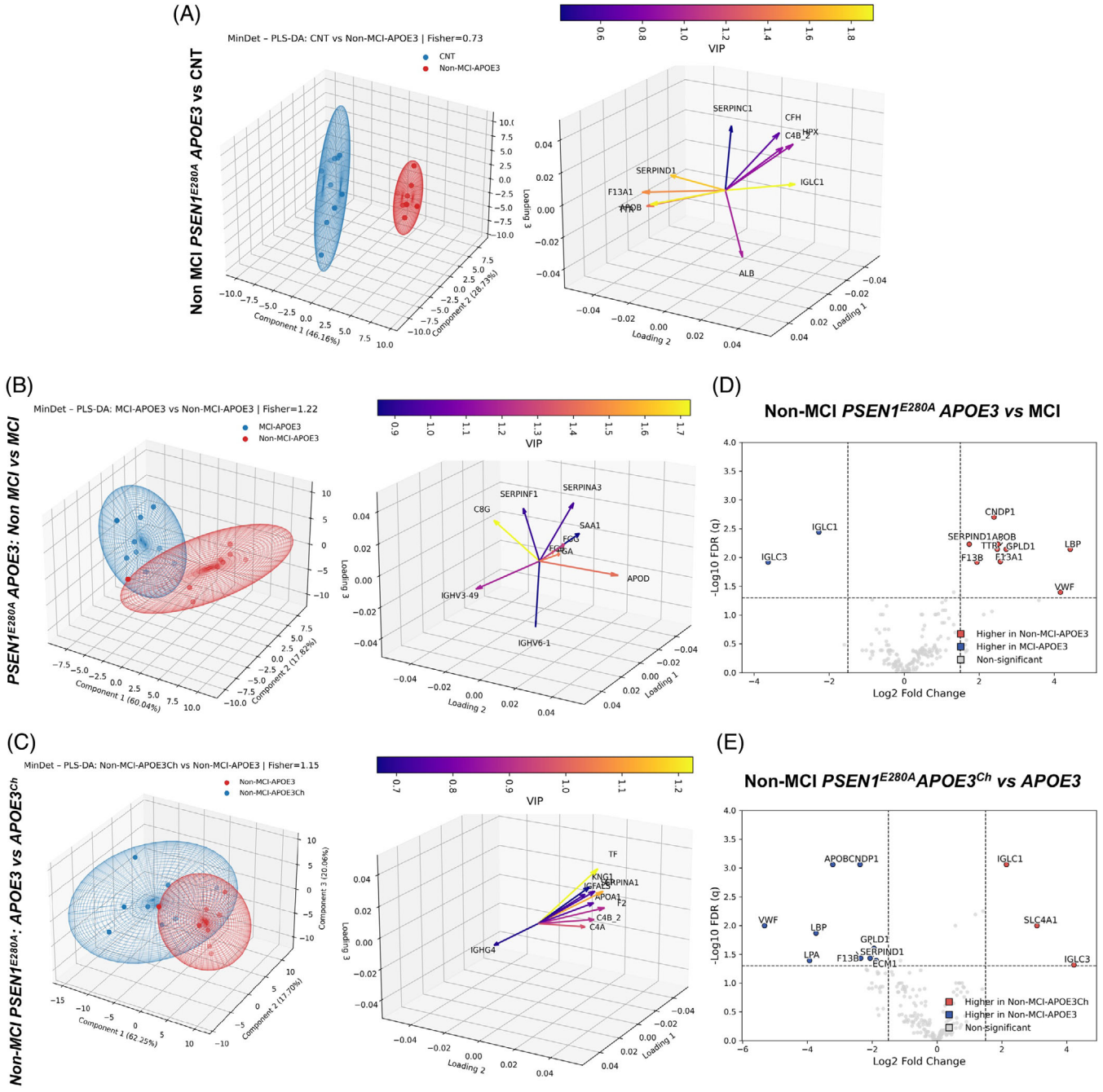


FIGURE 5 *APOE3*^{Ch} reduces vascular, inflammatory, and coagulation proteins in EVs from non-MCI *PSEN1*^{E280A} carriers. Partial least squares–discriminant analysis (left) and their respective loading weights (right) for (A) non-MCI *PSEN1*^{E280A} carriers and CNT individuals, (B) *PSEN1*^{E280A}-*APOE3* carriers with and without MCI, (C) non-MCI *PSEN1*^{E280A} carriers with *APOE3* and *APOE3*^{Ch}. D, Volcano plot showing differentially enriched proteins between *PSEN1*^{E280A} non-MCI and MCI. Red dots represent significantly enriched protein in the non-MCI group compared to MCI. E, Volcano plot showing differentially enriched proteins between *PSEN1*^{E280A} carriers with *APOE3* and *APOE3*^{Ch} carriers (both non-MCI). Blue dots represent significantly downregulated in the *APOE3*^{Ch} group compared to *APOE3* ($|\log_2$ fold change$| \geq 1.5$, P value < 0.05). Representative data from MCI *PSEN1*^{E280A}-*APOE3*, $n = 9$ (MCI1, MCI2, MCI3, MCI4, MCI5, MCI6, MCI7, MCI9, MCI10); non-MCI *PSEN1*^{E280A}-*APOE3*, $n = 9$ (non-MCI-1, non-MCI-2, non-MCI-3, non-MCI-4, non-MCI-5, non-MCI-7, non-MCI-8, non-MCI-9, non-MCI-10), and non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch}, $n = 9$ (E-Ch1, E-Ch2, E-Ch3, E-Ch4, E-Ch5, E-Ch6, E-Ch7, E-Ch8, E-Ch9). *APOE*, apolipoprotein E; CNT, cognitively unimpaired control; EV, extracellular vesicle; MCI, mild cognitive impairment; PSEN, presenilin

detectable DEPs; therefore, adjusted values are used as primary results, and unadjusted values are reported as exploratory.

In the non-MCI *PSEN1*^{E280A}-*APOE3* versus CNT comparison, the uncorrected analysis revealed increased inflammatory and vascular proteins (APCS, Von Willebrand factor [VWF], glycosylphosphatidylinositol specific phospholipase D1 [GPLD1], CNDP1, APOB, fibronectin 1 [FN1], IGHV3-64D), none of which remained significant after FDR correction (Table S7A,B in supporting information). In the MCI versus non-MCI *PSEN1*^{E280A}-*APOE3* comparison, the uncorrected analysis identified increases in LBP, VWF, lipoprotein A [LPA], GPLD1, coagulation factor XIII A chain [F13A1], APOB, transthyretin [TTR], CNDP1, and APCS. After FDR correction, most remained significant except LPA, SERPINA4, and APCS (Figure 5D, Table S7A,B). In the non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch} versus *APOE3* comparison, 23 uncorrected DEPs showed reduced abundance in *APOE3*^{Ch}-EVs, including VWF, LBP, GPLD1, APOB, CNDP1, F13B, SERPIND1, LPA, and extracellular matrix protein 1 (ECM1), while FDR correction retained 12 significantly decreased proteins. Three proteins, IGLC1, solute carrier family 4 member 1 (SLC4A1), and IGLC3, remained consistently enriched in *APOE3*^{Ch}-EVs across analyses, supporting roles in immunomodulatory or homeostatic functions (Figure 5E, Table S7A,B).

Overall, these DEP signatures show early vascular-inflammatory activation in *PSEN1*^{E280A}-*APOE3* carriers, whereas *APOE3*^{Ch}-EVs display the opposite pattern, supporting a distinct protective molecular profile.

3.6 | EV proteomic signatures in *APOE3*^{Ch} carriers suggest reduced vascular activation and inflammation associated with β-catenin signaling pathway

We next analyzed DEPs from non-MCI *PSEN1*^{E280A}-EVs to identify enriched canonical pathways using IPA. We generated a bubble plot of pathways in *PSEN1*^{E280A} carriers, highlighting predicted or significant pathway activity. In the non-MCI versus MCI comparison, the non-MCI group showed activation of the clotting cascade, acute phase response signaling, and serotonin receptor signaling. Pathways linked to vascular inflammation, the coagulation cascade, and cholesterol metabolism exhibited robust significance. We then compared these results to those from *APOE3*^{Ch} carriers. Interestingly, most of the pathways predicted to be activated or showing significance in non-MCI *APOE3* carriers appeared to be downregulated in non-MCI *APOE3*^{Ch} carriers, based on the profiles of the identified DEPs (Figure 6A). These included the acute phase response signaling, atherosclerosis signaling, response to elevated platelet cytosolic Ca^{2+}, and production of reactive oxygen species, all of which are processes implicated in vascular inflammation, coagulopathy, and neurodegeneration. In contrast, two pathways, interleukin (IL)-12 signaling and gamma interferon inhibitor of translation translational signaling, were upregulated in *APOE3*^{Ch} compared to *APOE3* (Figure 6A). This shift suggests a potential dampening of pathogenic inflammation signaling in *APOE3*^{Ch} carriers, consistent with their observed clinical resilience and vascular protection.

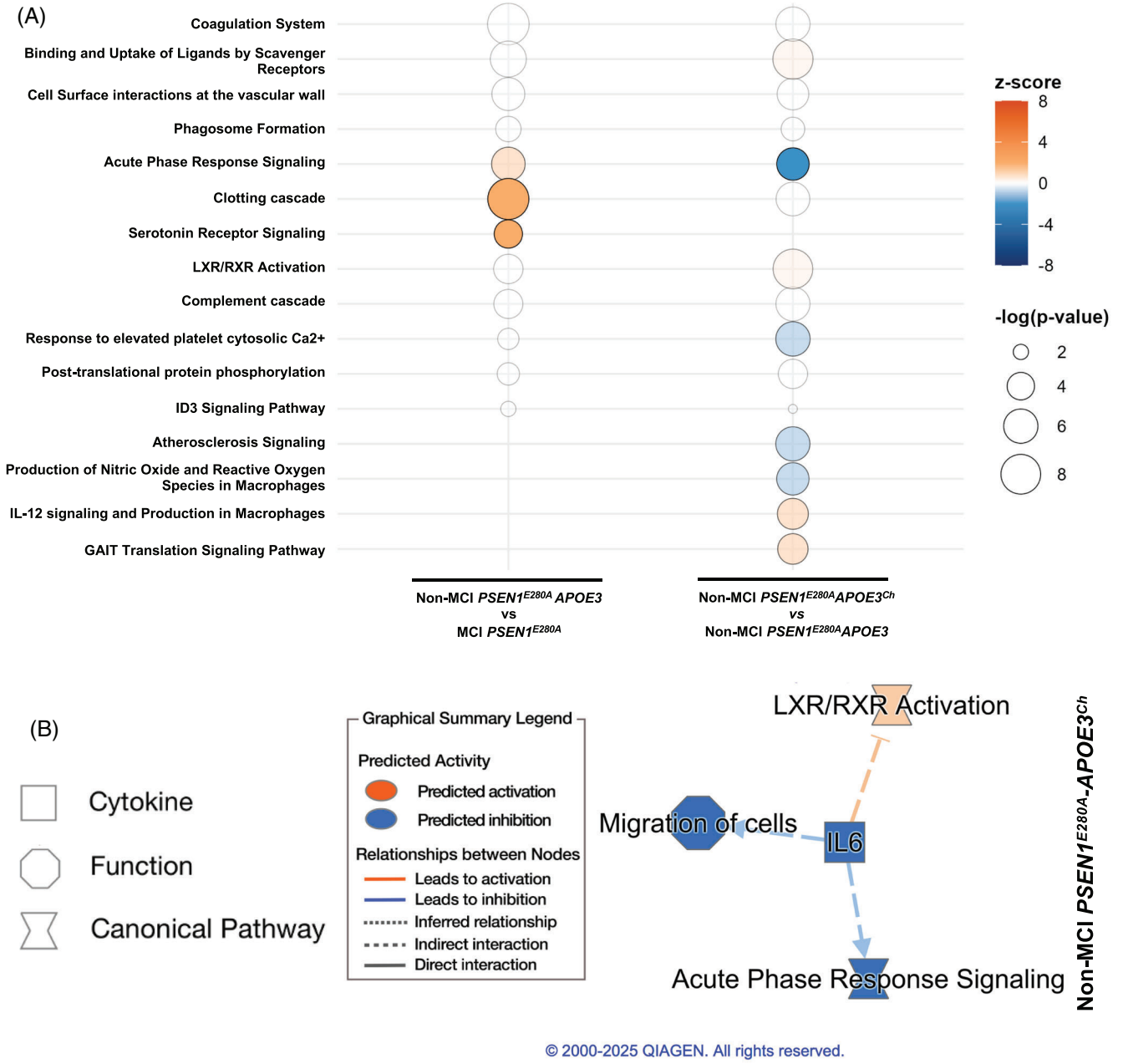
To further explore the molecular mechanisms underlying the observed pathway differences, we analyzed upstream regulators and predicted biological effects using IPA. In non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch}, IL-6 was predicted to be inhibited, resulting in reduced cell migration and acute-phase response signaling, as well as decreased activation of the liver X receptor/retinoid X receptor (LXR/RXR) pathway (Figure 6B).

IPA analysis revealed that non-MCI *PSEN1*^{E280A}-*APOE*3 versus MCI *PSEN1*^{E280A}-*APOE3* showed predicted activation of IL-6, linked to at least three DEPs involved in inflammation and lipid metabolism, indicating a shift toward a pro-inflammatory and metabolically dysregulated state. In contrast, non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch} exhibited the opposite pattern, with predicted inhibition of IL-6, CCAAT/enhancer-binding protein beta, and transforming growth beta factor 1, and additional upstream regulators (*APOE*, IL-2, APP, IL-1β, tumor necrosis factor) meeting the ≥ 3-target criterion. Uncorrected analysis also identified CTNNB1 (β-catenin) as an upstream regulator in *APOE3*^{Ch}, targeting *APOM*, *ATP*, *ECM1*, *F13A1*, and *FN1* genes linked to lipid metabolism, extracellular matrix remodeling, coagulation, and inflammation (Table S8A,B in supporting information). Together, these results indicate that *APOE3*^{Ch} carriers display a reversal of the inflammatory and lipid-metabolic signatures observed in non-MCI *PSEN1*^{E280A}-*APOE3*.

These findings suggest that, based on EV composition, *APOE3*^{Ch} may modulate vascular signaling by repressing pro-inflammatory and pro-migratory pathways while promoting molecular changes that reduce endothelial activation and adhesion.

3.7 | *APOE3*^{Ch} protected endothelial cell integrity modulating β-catenin localization

We conducted an in-depth analysis of transcriptomic data available from the *post mortem* prefrontal cortex of individuals affected by *PSEN1*^{E280A} and resistants carrying *APOE3*^{Ch}. This analysis incorporated a total of 81,440 single-nucleus transcriptomes derived from tissue samples collected from four *PSEN1*^{E280A}-*APOE3*^{Ch} patients, six *PSEN1*^{E280A}-*APOE3*, and six neurologically healthy controls (Figure 7A). Using canonical markers, we identified thirteen distinct cell populations in the prefrontal cortex. These included six excitatory neuron subtypes (EN: L2/3, L4, L5/6, NEFL, SEMA3A, 5-HT), two inhibitory neuron subtypes (IN: CGE and MGE), astrocytes (AQP4), oligodendrocytes (PLP1), oligodendrocyte precursor cells (LHFPL3), microglia (LRMDA), and vascular (CLDN5) cells—including endothelial (VWF) and mural cells (PDGFRB) (Figure 7B). We analyzed the Wnt/β-catenin signaling ligand, receptors, adapters, transcription factors, and target genes profile in the vascular and astrocytes. We found upregulation of Wnt/β-catenin genes in astrocytes and vascular cells, specifically in LFE1 (Figure 7C). These findings are consistent with prior observations in isogenic induced pluripotent stem cell (iPSC)-derived brain organoids, where the *APOE3*^{Ch} variant enhanced Wnt3a signaling and increased expression of β-catenin target genes, despite exhibiting reduced levels of Wnt ligands.[8] To investigate the effect of *APOE3*^{Ch}



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FIGURE 6 EV proteomics in *APOE3*^{Ch} carriers suggest reduced vascular activation and inflammation. A, Bubble plot of canonical pathways differentially activated in non-MCI *PSEN1*^{E280A} carriers with *APOE3* and *APOE3*^{Ch}, as identified by IPA. Bubble size represents –log(*P* value). From left to right, the columns represent the following group comparisons: non-MCI versus MCI, and non-MCI *APOE3*^{Ch} versus non-MCI *APOE3*. Color reflects IPA *z* score: white denotes no available activity pattern, orange indicates predicted pathway activation, and blue indicates predicted inhibition. B, Predicted regulators and molecular mediators in non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch} carriers. The panel shows regulators predicted to be inhibited (blue). *APOE*, apolipoprotein E; EV, extracellular vesicle; GAIT, gamma interferon inhibitor of translation; IL, interleukin; IPA, Ingenuity Pathway Analysis; LXR/RXR, liver X receptor/retinoid X receptor; MCI, mild cognitive impairment; PSEN, presenilin

on Wnt signaling and endothelial function, we treated HBMECs with recombinant ApoE3Ch protein. Both wild-type ApoE3 and ApoE3Ch reduced VEGF-induced vascular permeability and helped preserve TEER (Figure 7D). Furthermore, ApoE3Ch promoted nuclear localization of β-catenin, in contrast to ApoE4, which led to cytosolic displacement of β-catenin and disruption of endothelial junctions. Notably, this ApoE4-induced phenotype was reversed by ApoE3Ch treatment (Figure 7E). Collectively, these results support Wnt/β-catenin signaling as a potential protective mechanism conferred by the *APOE3*^{Ch} variant.

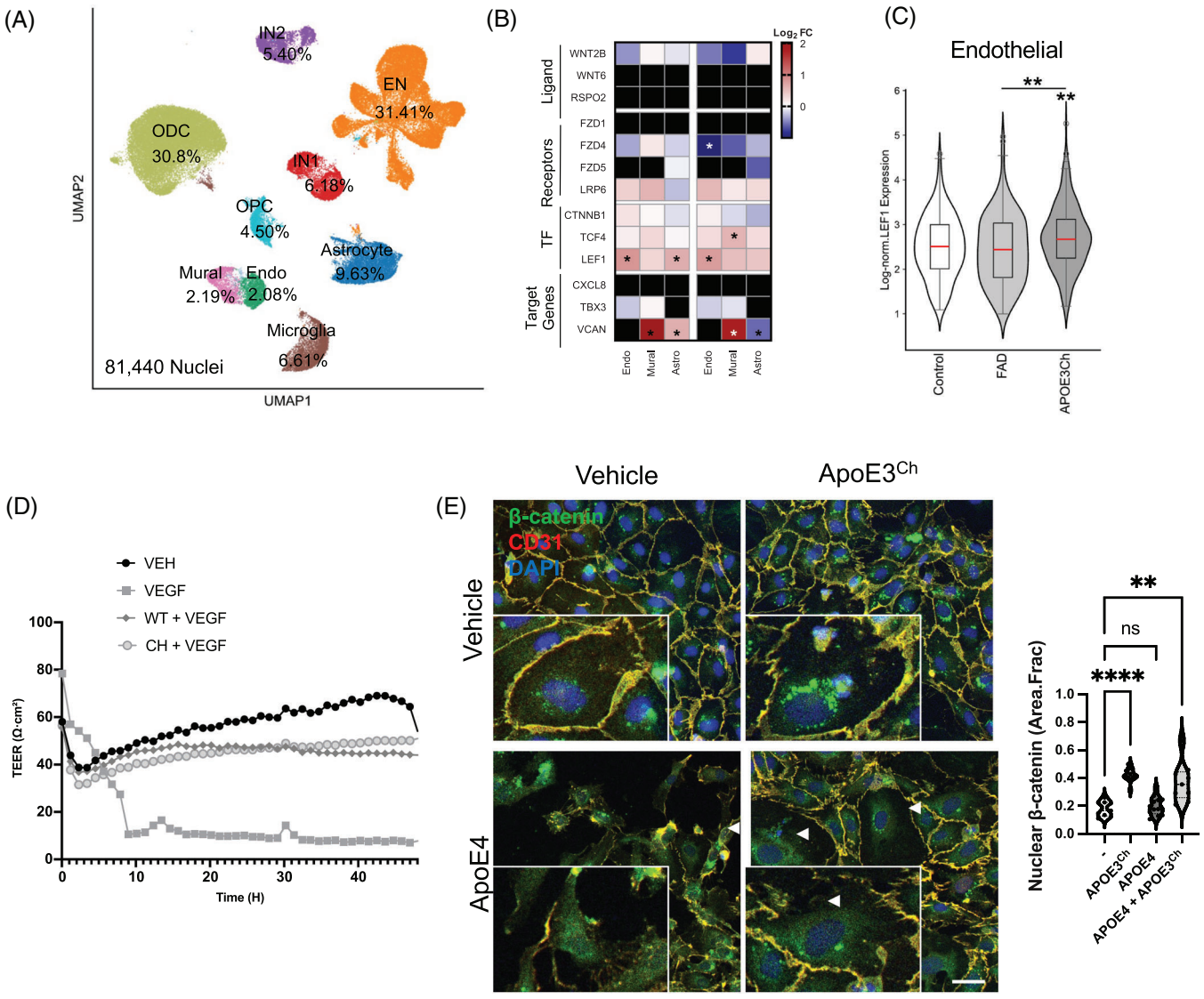


FIGURE 7 Wnt signaling expression in *APOE3*^{Ch} across vascular cells and conditions of AD, and its role in promoting vascular integrity. A, UMAP visualization of 81,440 cells from *post mortem* human prefrontal cortex. B, Heatmap of relative expression of canonical Wnt signaling mediators across vascular cell types in *APOE3*^{Ch} and non-carries with *PSEN1*^{E280A}, showing a significant increase WNT TF in endothelial cells. Significance showed for each gene and cell type (* $P < 0.05$). C, Violin plot of lymphoid enhancer-binding factor-1 expression in endothelial for non-disease (Control), FAD (*PSEN1*^{E280A}), APOE3Ch (*PSEN1*^{E280A} *APOE3*^{Ch}). D, TEER measurements in HBMECs showed that treatment with VEGF (50 ng/mL) led to a progressive reduction in TEER over a 48-hour period, reflecting increased endothelial permeability. However, this VEGF-induced decrease in barrier integrity was prevented when cells were co-treated with either wild-type ApoE3 or the ApoE3Ch variant at 1 μM concentration. E, On the left, representative images show β-catenin (green), CD31 (red), and DAPI (blue) in HBMECs co-cultured with astrocytes, treated with ApoE3Ch (1 μM) in combination with ApoE4 (1 μM). On the right, quantification of nuclear β-catenin demonstrates that ApoE3Ch enhances β-catenin nuclear translocation and restores adherens junction integrity disrupted by ApoE4. AD, Alzheimer's disease; *APOE*, apolipoprotein E; FAD, familial Alzheimer's disease; HBMEC, human brain microvascular endothelial cell; MCI, mild cognitive impairment; PSEN, presenilin; TEER, transendothelial electrical resistance; UMAP, Uniform Manifold Approximation and Projection; VEGF, vascular endothelial growth factor

3.8 | *APOE3*^{Ch}-EVs reduced GFAP and protected vascular cell integrity modulating β-catenin localization on cell dependent manner

We demonstrated that systemic EVs interact directly with endothelial cells and can cross the blood–brain barrier (BBB) to reach the brain parenchyma.[24] Building on this, we analyzed the impact of EVs on glia-endothelial activation and β-catenin localization in an astrocyte–endothelium co-culture model.

EVs from representative samples were FM4-64–labeled and internalized by recipient cells within 12 hours (Figure S4A). EV exposure

induced average cytotoxicity of 11% to 23% across groups (Figure S4B).

The non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch} EVs significantly reduced both the number and area of endothelial gaps and increased nuclear β-catenin localization compared to EVs from non-MCI *PSEN1*^{E280A}-*APOE3* carriers (Figures 8A,B and S5C,E in supporting information). Similarly, in astrocytes, *APOE3*^{Ch}-EVs reduced GFAP intensity but also the nuclear β-catenin fraction (Figures 8C,D, S5D,F). Given that EVs from *APOE3*^{Ch} carriers are more enriched in apoE (Figure 4A), we assessed whether this apoE was taken up by recipient cells. While it is not possible to determine the exact proportion of intracellular apoE that is EV-derived versus endogenous, both endothelial and astrocytic cells treated with *APOE3*^{Ch}-EVs tended to exhibit a higher number of intracellular apoE-positive clusters compared to those treated with EVs from *APOE3* carriers (Figures 8E,F, S5G and S5H), suggesting enhanced cellular uptake or accumulation of apoE in response to the EV treatment.

These findings suggest that *PSEN1*^{E280A}-*APOE3*^{Ch}, unlike *PSEN1*^{E280A}-APOE3-EVs, may promote endothelial integrity and astrocytic homeostasis, potentially through modulation of β-catenin signaling.

4 | DISCUSSION

This study provides the first characterization of plasma-derived EVs from cognitively unimpaired *APOE3*^{Ch} carriers, with or without the *PSEN1*^{E280A} mutation, compared to *PSEN1*^{E280A}-*APOE3* carriers with and without MCI. Non-MCI *PSEN1*^{E280A}-*APOE3* EVs showed elevated vascular and leukocyte markers correlating with poorer cognitive scores, alongside proteomic enrichment of proinflammatory and vascular pathways and functional evidence of endothelial disruption and astrocyte reactivity. In contrast, *PSEN1*^{E280A}-*APOE3*^{Ch} EVs displayed reduced vascular and leukocyte markers, but were enriched in apoE and neutral lipids, with proteomic signatures opposite to *PSEN1*^{E280A}-*APOE3* EVs, preserving endothelial integrity and attenuating astrocytic responses. Together, these findings suggest that *APOE3*^{Ch} reshapes EV composition toward a phenotype linked to vascular and glial protection.

The *PSEN1*^{E280A} mutation in the Paisa kindred provides a unique model for investigating the temporal trajectory of AD biomarkers across the preclinical and symptomatic stages. Classical biomarkers such as cerebrospinal fluid (CSF) and plasma define disease staging decades before onset but mainly reflect downstream protein pathology.[42] EVs, by contrast, may capture upstream molecular signals.[43] While studied mainly for carrying phosphorylated tau and propagating pathology,[21,22,44] our findings show that EVs also reflect inflammation, vascular dysfunction, and lipid metabolism changes before MCI, supporting their utility as early biomarkers of disease resistance or progression, particularly because EV markers including CD45, CD105, and PDGFRβ correlate with cognitive performance.

EV proteomic alterations were observed specifically in non-MCI individuals compared to both CNT and MCI groups, suggesting sensitivity to early pathophysiological changes preceding cognitive decline. Prominent signatures included inflammatory (LBP, SERPIN4, APCS, SERPIND1) and vascular activation pathways (VWF, FN1, LPA, APOB), commonly seen in cardiovascular and autoimmune disorders,[45,46] aligning with the enrichment of CD45, CD105, and CD45^{+}CD105^{+} EVs. In *APOE3*^{Ch}, IPA analysis indicated activation of anti-angiogenic IL-12 signaling,[47] along with predicted inactivation of IL-6, an upstream regulator associated with inflammation, vascular activation, and coagulation.[48] Distinctively in AD, this profile was associated with NVU-related markers,[9] indicating that EVs may mediate or reflect brain–periphery communication relevant to early neurodegeneration.

The EV profile in non-MCI *PSEN1*^{E280A}-*APOE3* carriers was reversed in *APOE3*^{Ch} carriers, suggesting suppression of early peripheral damage signals. Despite broader age variability, evidence supports a protective role for *APOE3*^{Ch}: homozygous carriers showed reversal of cerebral hypometabolism,[5] and heterozygotes exhibited reduced angiogenesis and cerebral amyloid angiopathy.[3] *Post mortem* studies showed preserved astrocytic architecture in *APOE3*^{Ch},[11] and astrocyte-derived EVs exhibited reduced AQP4, consistent with maintained BBB integrity.[49] These findings suggest less BBB damage in preclinical *APOE3*^{Ch} carriers, reflected in their EV profiles.

In our study, non-MCI *PSEN1*^{E280A}-*APOE3*^{Ch} carriers showed increased CD41a^{+} EVs, while IPA predicted coagulation activation in *PSEN1*^{E280A}-*APOE3* but inhibition in *PSEN1*^{E280A}-*APOE3*^{Ch}. Given that platelet EVs can also carry regulatory and pro-neurogenic factors,[50–52] this profile may reflect an altered platelet state favoring vesicle-mediated communication over thrombosis.

Our EV isolation protocol did not fully remove plasma components, with ≈ 15% lipoproteins remaining in the pellet. Some lipoproteins, often considered contaminants, may bind to EV membranes and persist after isolation[53,54] so the enrichment of apoE and neutral lipids in *PSEN1*$^{E280A-}$*APOE3*^{Ch} EVs should be interpreted cautiously. ApoE, mainly a plasma lipoprotein component,[55] can also exist in non-lipidated form and within EVs.[56,57] *APOE* variants contribute to lipid disorders[58]; like *APOE2*, *APOE3*^{Ch} may promote hyperlipidemia due to reduced receptor binding but also limit uptake of oxidized lipoproteins, protecting against lipid overload and oxidative stress.[7] In our study, *APOE3*^{Ch} carriers showed increased enriched in neutral lipids, likely cholesterol esters, abundant in both lipoproteins and EVs.[59] These may reflect a compensatory mechanism to export excess cholesterol and preserve NVU homeostasis, as cholesterol buildup in neurons and glia promotes stress and inflammation, whereas its removal may reduce lipid burden.[7,60] Further lipidomic and functional studies are needed to define the composition and protective role of these neutral lipid–enriched in AD.

Several upstream mechanisms may contribute to the protective phenotype associated with *APOE3*^{Ch}-EVs. Recent work shows that ApoE3Ch modulates Wnt pathway sensitivity by altering interactions with tau and reducing responsiveness to the Wnt inhibitor DKK1,[61] a mechanism also observed in *APOE3*^{Ch} iPSC-derived organoids that exhibit reduced tau pathology.[8] Beyond apoE, *APOE3*^{Ch}-EVs were enriched in IGLC1, IGLC3, and SLC4A1—proteins with potential immunomodulatory or homeostatic functions. Of note, SLC4A1 regu-

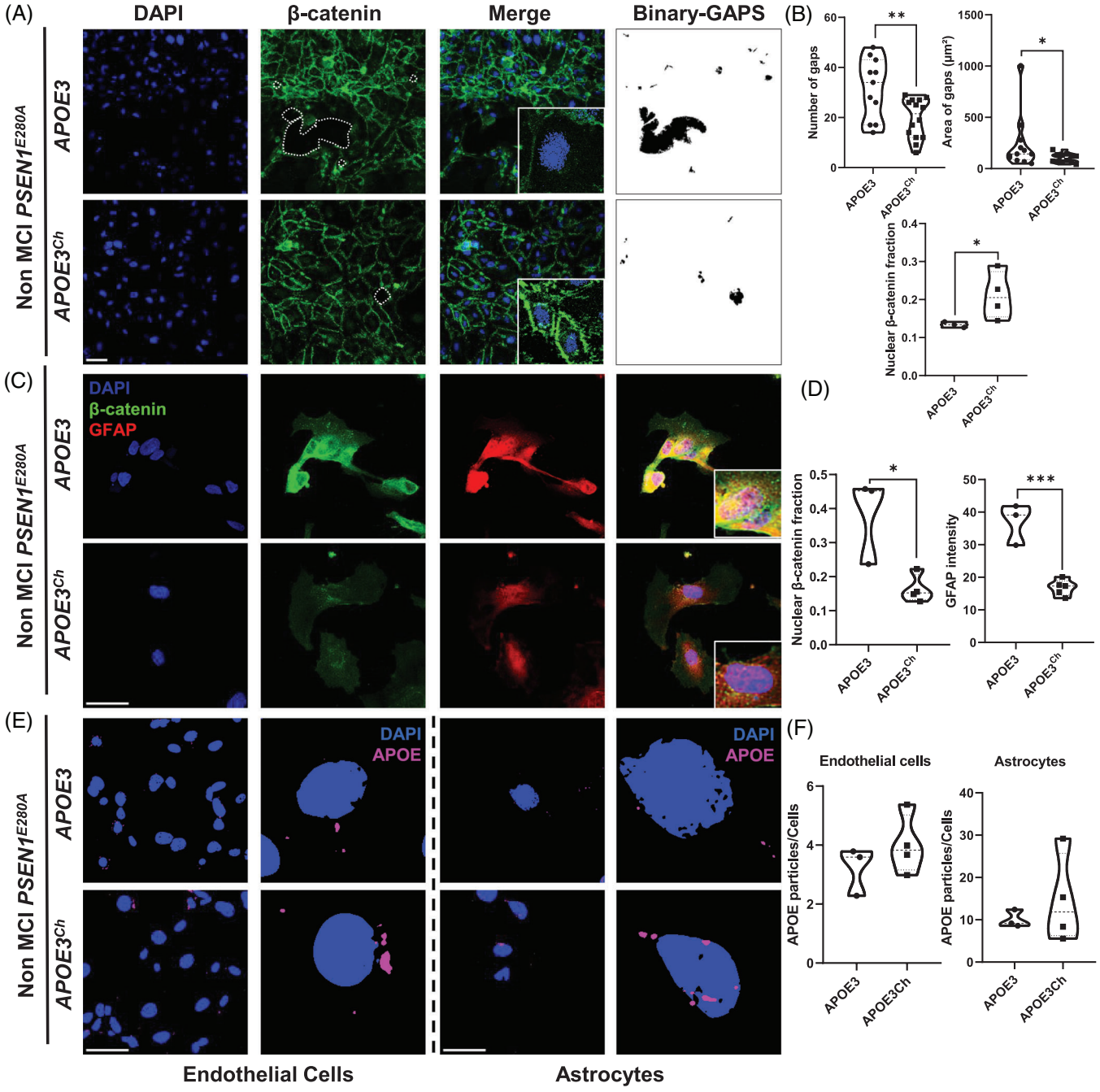


FIGURE 8 *PSEN1*^{E280A}-*APOE3*^{Ch} EVs preserve gliovascular integrity in an in vitro coculture model. A, Representative images of endothelial cells treated with EVs from *PSEN1*^{E280A} carriers with *APOE3* and *APOE3*^{Ch}. B, Quantification shows the number and area of intercellular gaps within the endothelial monolayer, as well as the fraction of nuclear *β*-catenin. C, Representative images of astrocytes treated with EVs. D, Quantification (right) includes GFAP intensity (red), reflecting astrocyte reactivity, and the fraction of nuclear *β*-catenin (green). E, Representative images showing ApoE signal localization in endothelial cells and astrocytes. F, Quantification depicts the number of ApoE-positive particles normalized to the total number of cells, highlighting differential ApoE uptake. Representative data from; non-MCI *PSEN1*^{E280A} -*APOE3*, $n = 4$ (non-MCI-2, non-MCI-3, non-MCI-6, non-MCI-7) and non-MCI *PSEN1*^{E280A} -*APOE3*^{Ch}, $n = 4$ (E-Ch6, E-Ch7, E-Ch8, E-Ch9). Magnification 20x and 40x, Scale bars: 50 µm. Violin plots illustrate the full data distribution, with individual data points overlaid and a line indicating the median. Unpaired *t* test, Mann Whitney test were performed. * indicates $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$. *APOE*, apolipoprotein E; EV, extracellular vesicle; GFAP, glial fibrillary acidic protein; MCI, mild cognitive impairment; PSEN, presenilin

lates peripheral pH and redox buffering, and its loss has been linked to oxidative stress and neurodegeneration,[62,63] suggesting that its enrichment in *APOE3^{Ch}*-EVs may support endothelial resilience. Additionally, the modest increase in lipid cargo in *APOE3^{Ch}*-EVs may reflect a regulated lipid-export mechanism. Consistently, *APOE3^{Ch}*-EVs proteomic signature show reduced LXR/RXR activation, and recent findings indicate that ApoE3Ch enhances the efflux of oxidized lipids, suggesting a coordinated shift in lipid handling.[64]

EVs from *PSEN1^{E280A}-APOE3^{Ch}* carriers preserved endothelial integrity and reduced astrocytic activation in co-culture. These effects may be linked to APOE3Ch cargo, consistent with activation of Wnt/β-catenin signaling in endothelial cells,[8] a pathway critical for BBB maintenance via regulation of tight junction proteins such as claudin-1.[65,66] Genetic inhibition of endothelial β-catenin exacerbates inflammation and barrier disruption, increasing CD4$^+$ T cell infiltration, VCAM-1 and caveolin-1 expression, and transcytosis in autoimmune encephalomyelitis models.[67] β-catenin also regulates PDGF-B, the ligand for PDGFRβ, promoting mural cell recruitment and BBB integrity,[68] a mechanism potentially underlying the vascular protection observed with *APOE3^{Ch}*-EVs. Supporting this, lymphoid enhancer-binding factor-1 was upregulated in endothelial cells, where it cooperizes with stabilized β-catenin to activate Wnt target genes.[69] Proteomic and IPA analyses also identified ID3, which enhances β-catenin transcriptional activity by blocking E47-mediated repression.[70,71]

β-catenin activation was not detected in astrocytes treated with *APOE3^{Ch}*-EVs, which may be beneficial because β-catenin signaling promotes astrocyte reactivity after injury.[72,73] Consistently, *APOE3^{Ch}*-EVs reduced GFAP expression, suggesting dampened astrocytic responses, possibly influenced by differential apoE uptake.[74] Consistently, *APOE3^{Ch}*-EVs reduced GFAP expression, suggesting dampened astrocytic responses, possibly influenced by differential apoE uptake.

We hypothesize that the protective phenotype in *APOE3^{Ch}* carriers is partially mediated by ApoE3Ch protein within EVs. ApoE has been detected in EVs and proposed as a therapeutic mediator,[56] while *APOE3^{Ch}* is linked to increased homeostatic astrocytes and microglia and reduced disease-associated populations.[75] ApoE, identified as an upstream regulator in *PSEN1^{E280A}-APOE3^{Ch}* vesicles, may indicate that vesicular ApoE3Ch modulates intracellular signaling pathways, including the activation of Wnt/β-catenin signaling,[8] which supports BBB integrity through junctional stabilization and transcriptional regulation[76] and limits immune infiltration.[67] Consistently, exogenous Wnt activation enhances endothelial barrier function,[77] suggesting that ApoE3Ch-EVs may act as modulators of this pathway.

Although our sample sizes (n = 9–10 per group) are appropriate for detecting the large effect sizes observed in EV and proteomic signatures, more subtle differences may remain underpowered and should be interpreted with caution. EV-based biomarkers also face important translational limitations. *APOE3^{Ch}* is extremely rare worldwide, and although Colombia contains the largest known affected population, larger cohorts and longitudinal sampling will be necessary for standardization. Our EV isolation strategy, based on differential centrifugation, is cost effective and aligns with the 2023 Minimal Information for Studies of Extracellular Vesicles guidelines,[19] but more specialized platforms (ultracentrifugation, affinity purification, Simoa) may increase analytical sensitivity. Reproducibility will also require larger cohorts and comparison across additional *APOE* genotypes, as small sample sizes limit the translational interpretation of EV biomarkers. This is underscored by recent work, which identified inflammatory signatures in *APOE* ε4/ε4 across brain, CSF, and plasma using high-powered cohorts.[78] Applying similar signature validation in *APOE3^{Ch}*-EVs will be essential to determine whether their lipid profiles reflect true protective metabolic adaptation or unmeasured confounders. Together, these considerations outline the key technical and biological steps needed before clinical translation.

In summary, plasma-derived EVs from cognitively unimpaired *PSEN1^{E280A}-APOE3* carriers reflect early pathological changes linked to inflammation, vascular alterations, and NVU dysfunction that disappear by the MCI stage, suggesting they are transient preclinical signals. These alterations were absent in *PSEN1^{E280A}-APOE3^{Ch}* carriers, supporting a protective role for the variant. Thus, EVs may serve as indicators of early mechanisms underlying both disease progression and resistance in genetically at-risk individuals.

AUTHOR CONTRIBUTIONS

L. P.-L.: methodology, investigation, analysis, visualization, writing draft, writing and editing; **D. A.**: conceptualization, methodology, investigation, analysis; **J. V.-V.**: methodology, conceptualization; **P.V.-C.**: methodology, conceptualization; **A.G.**: methodology, conceptualization; **A. U.**: analysis, visualization; **S. A-A.**: investigation; **L. C.**: investigation; **K. M.-C.**: Investigation; **D. C.**: conceptualization, methodology; **G. P. C-G**: conceptualization; **G. J. F.**: investigation; **L.A.K**: investigation; **J. F. A.-V.**: writing and editing, supervision, funding; **R. P.-D.**: conceptualization, methodology, investigation, analysis, visualization, writing and editing, supervision.

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CONFLICT OF INTEREST STATEMENT

Dr. J. F. Arboleda-Velasquez and Leo A. Kim are listed as co-inventors on a patent to leverage therapeutics based on the ApoE3 Christchurch findings filed by Mass General Brigham. All other authors have no conflicts to disclose. Author disclosures are available in the supporting information.

CONSENT STATEMENT

The study was approved by the institutional review boards of the University of Antioquia, including the Comité de Bioética de Investigación en Humanos (Acta 119 del 14 de agosto de 2018). All participants provided written informed consent for the publication of any potentially identifiable data or images included in this article.

DATA AVAILABILITY STATEMENT

The mass spectrometry proteomics data have been deposited to the ProteomeXchange Consortium via the PRIDE[79] partner repository with the dataset identifier PXD068363.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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